Vertical Competition on a Common Platform

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We consider a platform whose infrastructure constitutes common attributes for vertically differentiated products in a digital goods market. We examine how vendors craft individual attributes of their products, and how the platform decides on the provision and pricing of its infrastructure when facing vendors' strategic reactions. First, we find that when the platform offers more powerful infrastructure, the high-quality vendor improves individual attributes, whereas its low-quality rival reduces individual attributes, exhibiting a pattern of divergence. On the other hand, when the platform charges a higher commission rate for using its infrastructure, both vendors slash individual attributes, with the high-quality vendor slashing more. Second, we find that when vendors' development cost declines but remains high, the platform improves infrastructure and reduces the commission rate. But when vendors' development cost is already low and keeps declining, the platform scales back in infrastructure but still reduces the commission rate. Finally, we show the strategic response of the platform, when coupled with vendors' improved product development efficiency, further boost the platform profit, both vendors' profits, and consumer surplus, leading to a win-win-win outcome.

Key words: common attributes, vertical competition, platform infrastructure, game theory

1. Introduction

Vendors competing to sell through a common platform face unique opportunities and challenges. On the one hand, the platform, typically a digital marketplace or service hub, provides common infrastructure that streamlines vendors' development process, ensures consistency in user interface, and so enhances user experience with vendors' products. For example, Steam.com, the largest online platform where users buy, play, create and discuss PC games, is known for its easiness of use.[1] As one of Steam users commented: *"You can add mods to your game with a simple click, and you (usually) don't have to toy with the game files to get them to work. No other desktop client makes it this easy to make, download, or install mods for your games."*[2] Apple's iOS supports a large number of app developers with tools and resources that facilitate the distribution and monetization of apps.[3] On the other hand, competing vendors operating on the shared ecosystem strive to differentiate. For instance, high-quality note-taking apps on iOS such as GoodNotes offer polished interfaces with advanced features such as handwriting recognition and cloud sync,[4] whereas lower-quality options, such as Note-Ify or Evernote, fall short in these areas.[5] Premium video games released on Steam.com such as Skyrim and Red Dead Redemption compete by providing a polished, immersive experience that their low-quality counterparts can hardly match. Such quality variance is also reflected in game ratings (from no review to 95% positive reviews).[6]

For competing vendors, the platform infrastructure provides the "common attributes" that are indispensable and inseparable from their products. Consumers' perceived value of a vendor product (e.g., a video game on Steam.com) not only stems from the product's individual attributes but also depends on the common infrastructure (e.g., platform reliability). Technology platforms

[1] `https://store.steampowered.com/about/`

[2] `https://whatnerd.com/why-steam-most-popular-pc-gaming-platform/`

[3] `https://shorturl.at/cJzPf`

[4] `https://www.goodnotes.com/features`

[5] `https://nira.com/evernote-history/`

[6] `https://gamalytic.com/steam-analytics`

are also known to invest heavily in enhancing their infrastructure. For example, Steam.com constantly announces updates in developer tools,[7] which help developers enhance game development, marketing, and user engagement. Because common attributes from the platform constitute an indispensable part of product values, their changes will shake the balance of competition in a vertically differentiated market. In particular, vendors that target the high-end market and the low-end market may take different reactions in adjusting individual attributes of their products, which have implications for their pricing, competition outcome, and consumer welfare. Yet changes in common attributes shared by horizontally differentiated products do not affect their relative values and so are unlikely to induce asymmetric responses from vendors. In this study, we theorize how common attributes provided by a platform affect vertical competition in a digital goods market.

In practice, a platform typically monetizes its infrastructure provision by charging vendors a transaction-based commission (Borck et al. 2020). For example, Steam.com takes a 30% commission on sales made through its platform, including video games, downloadable content, or other virtual goods sold within games.[8] The commission can be viewed as the price vendors pay for using the infrastructure. Naturally, a platform's commission rate will interact with its infrastructure provision to affect vendor competition.

Moreover, a strategic platform can benefit from adjusting the provision and pricing of common infrastructure when facing changes in vendor productivity. For instance, third-party GenAI tools such as Promethean AI, Meshy.ai, and Rosebud AI can help game developers enhance gameplay mechanics, improve storytelling and animation, and streamline development processes,[9] lifting productivity for all vendors on Steam.com. Such universal improvement in vendor productivity reshapes vendors' competition as well as the platform's strategy to profit from the vendor market.

Aiming at understanding the strategic implications of a platform's common infrastructure in a vertically differentiated market, we ask the following research questions: 1) How would high-

[7] `https://partner.steamgames.com`

[8] `https://www.ncesc.com/gaming-pedia/what-percentage-cut-does-steam-take/`

[9] `https://www.analyticsvidhya.com/blog/2025/01/genai-in-video-game-development/`

and low-quality vendors adjust their products' individual attributes and pricing, in response to advancement in platform infrastructure or a higher commission rate for using the platform? 2) Considering vendors' strategic reactions, how should a platform decide its offering of common infrastructure and the corresponding commission rate? 3) Should the platform expand or scale back its infrastructure to take advantage of enhanced productivity of vendors, and how would it adjust the corresponding commission rate?

To address these research questions, we model a platform whose infrastructure (e.g., user interface, reliable payment, and order fulfillment) constitutes common attributes for vertically differentiated products in a digital goods market. The platform collects a commission for each unit of vendor product sales. After the platform sets the infrastructure and commission rate, competing vendors decide individual attributes as well as product prices to maximize profit.

Our investigation reveals how advances in platform infrastructure affect vendors' vertical competition. Conventional wisdom suggests that vendors will slash individual attributes to take advantage of the enhancement in common attributes, a *substitution effect*. Surprisingly, we find that although the low-quality vendor reduces individual attributes, the high-quality vendor in this case increases its individual attributes instead. This happens because more desirable common attributes exert a "bottom-up" influence, with a stronger impact on the low-quality vendor than on the high-quality vendor. The competition status of the low-quality vendor is strengthened as a result, providing incentive for the high-quality vendor to position away from its rival by elevating its individual attributes. We find that for digital goods with trivial marginal cost, this incentive is strong enough to dominate the substitution effect.[10] On the other hand, a higher commission rate discourages both vendors from offering individual attributes in a "top-down" pattern. In particular, the high-quality vendor reduces individual attributes in a greater magnitude than its low-quality rival. Accordingly, both vendors slash their prices in spite of the higher commission rate, and sell to more consumers.

We then investigate the platform's decisions related to the provision and pricing (i.e., the commission rate) of its infrastructure when considering vendors' strategic reactions. Intuitively, the

[10] For products with substantial marginal cost, the substitution effect may dominate (proof available upon request).

platform will offer better infrastructure if doing so becomes more cost-effective. We thus focus our investigation on an interesting scenario when technology advancements in the vendors' industry reduce their product development costs and consequently incentivize them to offer better individual attributes. We find that when vendors' development cost becomes lower, the platform first improves, and then reduces its infrastructure, following a non-monotonic pattern. Nonetheless, the platform lowers its commission rate always. These strategic activities benefit the platform by influencing the vendors' incentive to enhance individual attributes. Specifically, the platform's responses boost the increase in individual attributes for the high-quality vendor when vendors' product development efficiency improves but remains low, and boost the increase in individual attributes for the low-quality vendor when such efficiency is already high.

Finally, we find the strategic response of the platform, when coupled with the technology advancement of vendors, leads to a win-win-win outcome for all market players. Not only does the platform improve profit from its strategic move, but both vendors also obtain greater profits and consumers obtain greater total surplus. This result highlights the critical role of technology platforms in supporting the digital economy as the providers of common infrastructure. Furthermore, we explore the scenario where the platform, in addition to collecting commission income, also benefits from inducing a higher sales volume. We find that a platform that receives mixed income always charges a *higher* commission rate, compared to the case when it receives commission income only.

In summary, this paper examines the strategic implications of platform infrastructure on vertical vendor competition in a digital goods market. We contribute by theorizing how common attributes affect competing vendors' quality and pricing decisions. Our analyses reveal the asymmetric responses of vertically differentiated firms[11] in their crafting of individual attributes when common attributes become more appealing, along with the strategic implications of firms' strategic reactions on market competition and consumer welfare. Understanding the impact of enhanced common attributes is particularly important in our research context of digital goods market, where

[11] Throughout the paper, we use the terms "firms" and "vendors" interchangeably.

the trivial marginal cost leads to fierce competition and allows the high-quality firm to enjoy dominating power (Jones and Mendelson 2011). As such, our results shed lights on how platform infrastructure shapes vertical competition in a digital goods market.

This paper also offers timely insights into today's market environment where firms increasingly depend on common infrastructure provided by powerful digital platforms for their operations. Such platforms help small vendors develop and distribute innovative products, which otherwise would be unable to survive due to the lack of resources for mass advertising. We show that facing improved product development efficiency in the vendor industry, a strategic platform benefits from tailoring its offering of common infrastructure and commission rate to support and boost vendors' incentive to enhance products' individual attributes. The resulting win-win-win outcome from the platform's strategic response points to how the growth of technology platforms helps fuel the digital economy.

The remainder of the paper is organized as follows. We review related literature in Section 2 and set up the model in Section 3. In Section 4, we analyze the subgame of vendor competition and the platform's strategies. In Section 5, we demonstrate the robustness of the key results, which helps deepen our understanding of the underlying drives of the main model. Section 6 concludes.

2. Literature Review

Our work is related to two streams of literature: common attributes in vertical competition and platform provision of infrastructure, which we discuss in turn.

Common Attributes in Vertical Competition

Early studies on common attributes have largely focused on monopoly settings where multiple products offered by a monopolist share common manufacturing technologies (e.g., both Lexus and Avalon ride on the same chassis). For example, researchers have examined product line configuration when adopting common components can lower manufacturing costs (Kim and Chhajed 2000, Desai et al. 2001, Krishnan and Gupta 2001, Heese and Swaminathan 2006). Research on common attributes shared by competing firms is scant, with the exception of Ke and Lin (2020). This work considers common attributes shared by horizontally differentiated products and examines

competing firms' pricing decisions when product qualities are exogenously given but unknown to consumers. In this setting, changes in common attributes have symmetric impacts on competing firms. In contrast, our study considers common attributes shared by vertically differentiated products and examines competing firms' quality as well as pricing decisions. In this vertical competition context, changes in common attributes exert a "bottom-up" influence, with a stronger impact on the low-quality firm than on the high-quality firm.

By investigating how common attributes affect vertical competition, our study contributes to the extensive literature on firms' quality choices under vertical differentiation (e.g., Mussa and Rosen 1978, Bonanno 1986, Gabszewicz et al. 1986). The majority of this research considers product markets where the firm incurs substantial marginal production cost (e.g., material, labor) (e.g., Shaked and Sutton 1982, Moorthy 1984, 1988, Chambers et al. 2006, Shugan et al. 2017). In contrast, our analysis focuses on digital goods markets characterized by trivial marginal cost, and we observe the interesting finding that the high-quality firm increases individual attributes in response to more desirable common attributes. Shugan et al. (2017) examines a monopoly firm's decision about whether to bundle a core product with an add-on in a product line when the quality of the core product and the add-on are both endogenously determined. We instead consider common attributes that are intrinsic and inseparable from the product (e.g., technology infrastructure).

A few existing works examine vertical competitive markets where the firm incurs negligible marginal production cost relative to development cost (e.g., Lehmann-Grube 1997, Wei and Nault 2014). The literature shows that when marginal cost is trivial, price discrimination is not profitable so that a monopolist will only offer one product instead of a product line (e.g., Anderson and Dana 2009, Bhargava and Choudhary 2008). Jones and Mendelson (2011) further considers duopoly competition and shows that trivial marginal cost leads to high level of market concentration, where the high-quality firm has dominating power over the low-quality firm. In contrast, in a regular goods market with substantial marginal cost, the high-quality and the low-quality firms have more balanced market power. Amacher et al. (2005) compares social welfare optimum and duopoly

equilibrium for markets with negligible or substantial marginal costs, respectively. Our study adds to this research by investigating how competing vendors devise individual attributes in response to common platform infrastructure.

Our study is related to the broad literature on information goods, which are known to have trivial marginal cost. The majority of this research examines demand-side characteristics of information goods, such as low cost of sharing (Varian 2000, Bakos et al. 1999, Galbreth et al. 2012) and prevalence of product renting (Sundararajan 2004, Balasubramanian et al. 2015, Choudhary 2010). Our study adds to the stream of research that examines supply-side characteristics of information goods, i.e., trivial marginal cost. For example, Bakos and Brynjolfsson (1999) and Bakos and Brynjolfsson (2000) consider how the trivial marginal cost of information goods can affect bundling strategies. We show that the trivial marginal cost plays a major role on how common attributes affect vertical competition.

Platform Provision of Infrastructure

Our study examines how a platform decides the provision and pricing of its infrastructure by considering their implications for vendor competition. A body of literature considers a two-sided platform's decision on its own performance ("quality") level that may affect user participation. Anderson et al. (2014) recognizes that, in cases such as video game platforms, even though superior platform performance such as improved graphics is a selling point for consumers, the level of technological complexity may increase game development costs and thus discourage the entry of game developers. Tan et al. (2020) examines the choice of direct investments that increase participation on both sides of the platform and interact with its two-sided pricing decisions. Casadesus-Masanell and Llanes (2015) studies incentives to invest in platform quality in open-source and proprietary two-sided platforms. Whereas this literature abstracts away vendor competition, we model competing vendors that decide individual attributes and pricing for vertically differentiated products in response to platform infrastructure and show how vendor's strategic reactions drive the platform's infrastructure provision and pricing strategies.

With a focus on vendor quality competition, our analysis is related to the emerging literature on quality regulation and infrastructure provision by platforms. To regulate vendors' quality, the platform could use an exclusion strategy (Hagiu 2009, Zheng and Kaiser 2013), a first-party applications strategy where the platform becomes vertically integrated (Hagiu and Spulber 2013, Lee 2013), and a subsidization strategy (Huang et al. 2022). The aforementioned works consider the quality of upstream vendors as exogenously given. In our study, vendors endogenously determine their individual attributes, and the common infrastructure provided by the platform is an integral part of the product value to consumers.

Broadly speaking, our study relates to the extensive research investigating how a major retailer induces strategic responses of upstream manufacturers. Researchers have investigated manufacturer strategies such as information sharing (Gal-Or et al. 2008), assortment (e.g., Dukes et al. 2009, Inderst and Shaffer 2007), product specification (e.g., Luo et al. 2007, Dukes et al. 2014), wholesale prices (Gu and Liu 2013), and retail prices (e.g., Chen 2003, Marx and Shaffer 2007, Raju and Zhang 2005). These studies typically consider a wholesale price contract between manufacturers and the big retailer, where the retailer benefits from intensified manufacturer competition. In contrast, our study considers a revenue-sharing scheme between the platform and vendors, in which case the platform's incentive to maximize commission income aligns with vendors' incentive to enhance sales revenue. Previous research has examined the interactions between the platform and vendors on a revenue-sharing contract, including information asymmetry (Bart et al. 2021), the nature of bargaining (MacDonald and Ryall 2004, Oh et al. 2015), how platforms squeeze the surplus from vendors (Balseiro et al. 2017), product recommendations (Li et al. 2018, 2020, Zhou and Zou 2023), and differential commission rates (Bhargava et al. 2022). Most studies on revenue sharing in platform context assume that vendor qualities are exogenously given (e.g., Jiang et al. 2011, Ryan et al. 2012, Abhishek et al. 2016, Ha et al. 2022). Hsiao et al. (2024) consider endogenous quality decisions of competing manufacturers who decide on their product quality *before* the platform chooses its commission rate.

Recent studies compare the revenue-sharing contract with other contracts on a retail platform. They use numerical simulations to explore endogenous quality decisions of competing manufacturers *after* the platform sets the commission rate (Zhang et al. 2019, Xiong et al. 2024). Differently, in our research context, the platform infrastructure constitutes the common attributes that are indispensable and inseparable from vendors' products, and the commission rate can be viewed as the price for using such infrastructure. We characterize how the platform *jointly* decides its infrastructure provision and commission rate, and how these decisions affect competing vendors in setting their individual attributes.

3. Model Setup

We consider a platform that provides infrastructure to two competing firms, each selling a single product to consumers.

Platform The platform's infrastructure constitutes common attributes of products offered by vendors. Let $E \geq 0$ denote the value of common attributes, a summary measure for all more-is-better attributes shared by both products. For example, online video game platform Steam.com provides a powerful infrastructure that allows for smooth experiences for users to make, download or install mods for their games. The platform experience together with a game's playing features constitutes consumers' perceived value of the game. This is consistent with the notion that a product sold online can be viewed as a bundle of the core product (e.g., the product's key features) and the extended product (e.g., service provided by the online seller) (Ba and Pavlou 2002, Dellarocas 2003). In such a market environment, changes in platform infrastructure may induce a vendor to alter its quality provision, which is similar to the practice where vendors alter their products for different sales channels.[12] The platform incurs a cost $\frac{\lambda E^2}{2}$ for providing E to vendors, where

[12] For example, Black & Decker sells 20 Volt Lithium-Ion Cordless Drill through both Walmart and Home Depot. The Home Depot one comes with a 1.5 Ah battery, whereas the Walmart one only comes with a 1 Ah battery (https://qr.ae/pvWrB5). For Rain-X windshield washer fluid sold through Target and Walmart, they look identical except that Walmart's only goes down to -20F instead of -25F, which is the normal range (https://qr.ae/pvWrQp).

$\lambda \geq 0$ is the cost coefficient. A higher value of λ indicates a higher cost for the platform to provide infrastructure.

The platform charges a commission for its provision of common infrastructure. Following the widely used revenue-sharing scheme (e.g., Chen et al. 2016, Bhargava et al. 2022), we assume that the platform collects a portion α of the total revenue and each vendor keeps $(1-\alpha)$ portion of its own revenue, $\alpha \in [0,1]$. For instance, Steam.com takes a 30% cut of the revenue generated from game sales, downloadable content, and in-game purchases.[13] We assume that α is industry-specific and thus the same for both vendors. If a vendor pays the platform for additional, customized services, such payment can be viewed as the vendor's cost of improving its individual attributes.

Vendors We consider a product equilibrium in which the vendors offer different levels of individual attributes q_H and q_L, respectively, with $q_H > q_L \geq 0$. That is, we look for an equilibrium in which the vendors offer different levels of individual attributes. Otherwise, if they choose the same individual attributes, fierce competition will force both firms to price at unit cost, leading to a dominated equilibrium where both firms earn negative profits (Jones and Mendelson 2011, Lin 2017). In our research context of digital goods, the marginal cost is typically trivial. We thus assume that for individual attributes q_i $(i = H, L)$, firm i incurs zero marginal cost and an overall development cost $F_i = \frac{1}{2} f q_i^2$, where $f \geq 0$ reflects the industry's production development technology. A smaller value of f indicates a lower cost for product development in the vendor industry. Common attributes E and individual attributes q_i together constitute the *overall quality* $Q_i = q_i + E$. Clearly, the quality gap $Q_H - Q_L$ is equivalent to the difference in individual attributes $q_H - q_L$.

Consumers A continuum of consumers differ in their marginal valuations for quality denoted as θ, which is uniformly distributed on $[0,1]$. θ is privately known by individual consumers, with the distribution being common knowledge. The entire market size is normalized to 1. Each consumer is interested in buying at most one unit of the product. We consider common attributes that are intrinsic, and indispensable to the product. Following the literature (e.g., Lin 2017, Balachander

[13] https://stepofweb.com/steams-cut-how-much-do-developers-lose/

et al. 2017), we assume a consumer has the same marginal valuation for common attributes and for the product's individual offering.[14] The net surplus for a type θ consumer who purchases one unit of product $i = H, L$ at price p_i is given by

$$u_i(\theta) = \theta(q_i + E) - p_i. \tag{1}$$

Consumers buy product i if and only if $u_i > u_{-i}$ and $u_i \geq 0$.

Game Sequence The game has three stages: In stage 1, the platform decides E and α. In stage 2, competing vendors simultaneously choose individual attributes q_H and q_L. In stage 3, both vendors simultaneously choose product prices p_H and p_L; consumers make purchase decisions.

4. Analysis

We solve the game backwards. §4.1 derives the subgame equilibrium of vendor competition, given the platform's infrastructure provision E and commission rate α. §4.2 then explores the platform's optimal strategies.

4.1. Subgame of Vendor Competition

In this subgame, each vendor decides its individual attributes and price, taking the platform's common attributes E and commission rate α as given. Let θ_{HL} denote the type of the threshold consumer who is indifferent between purchasing product L and product H, i.e., $\theta_{HL}(q_H + E) - p_H = \theta_{HL}(q_L + E) - p_L$. Similarly, θ_{L0} denotes the type of threshold consumer who is indifferent between purchasing product L and not purchasing at all, i.e., $\theta_{L0}(q_L + E) - p_L = 0$. From the two equations, we obtain:

$$\theta_{HL} = \frac{p_H - p_L}{q_H - q_L}, \tag{2}$$

$$\theta_{L0} = \frac{p_L}{q_L + E}. \tag{3}$$

[14] Prior literature on two-dimensional quality competition has modeled the situation where consumers have independent willingness to pay for different product attributes (e.g., Vandenbosch and Weinberg 1995, Lauga and Ofek 2011). Their focus is on how to best leverage differentiation in product attributes for positioning. In our paper, we focus on how common attributes affect firms' choice of individual attributes when consumers have a holistic preference for the product.

Demands for firm H and firm L are given by $D_H = 1 - \theta_{HL}$ and $D_L = \theta_{HL} - \theta_{L0}$, respectively. The profit functions of the two firms are thus given by:

$$\pi_H = (1-\alpha)p_H(1-\theta_{HL}) - \frac{f}{2}q_H^2, \tag{4}$$

$$\pi_L = (1-\alpha)p_L(\theta_{HL}-\theta_{L0}) - \frac{f}{2}q_L^2. \tag{5}$$

In stage 3, vendors' simultaneously choose product prices, taking individual attributes as given. Given q_H and q_L, we use first-order conditions to obtain the vendors' equilibrium prices:

$$p_H(q_H, q_L) = \frac{2(q_H+E)(q_H-q_L)}{(4q_H-q_L+3E)}, \tag{6}$$

$$p_L(q_H, q_L) = \frac{(q_L+E)(q_H-q_L)}{(4q_H-q_L+3E)}. \tag{7}$$

The price gap is given by

$$p_H(q_H, q_L) - p_L(q_H, q_L) = \frac{2(q_H-q_L)(2q_H+E-q_L)}{2(4q_H-q_L+3E)}, \tag{8}$$

which decreases with E. Plugging equations (6) and (7) back to the vendors' profit functions, we solve for their equilibrium individual attributes in stage 2, as characterized in the following lemma.

LEMMA 1 **(Equilibrium individual attributes)**. *Given platform's provision of infrastructure E and commission rate α, vendors' equilibrium individual attributes $q_H^*(E,\alpha)$ and $q_L^*(E,\alpha)$ satisfy the first order conditions as follows:*

$$(1-\alpha)(q_H+E)^2(4q_H-7q_L-3E) - fq_L(4q_H-q_L+3E)^3 = 0, \tag{9}$$

$$4(1-\alpha)(q_H+E)\left(2(q_L+E)^2 - 3(q_L+E)(q_H+E) + 4(q_H+E)^2\right) - fq_H(4q_H-q_L+3E)^3 = 0. \tag{10}$$

The equilibrium prices of the two products $p_H^*(E,\alpha)$ and $p_L^*(E,\alpha)$ can be obtained by plugging back $q_H^*(E,\alpha)$ and $q_L^*(E,\alpha)$ to equations (6) and (7). Vendors' equilibrium profits can be obtained by plugging $q_i(E,\alpha)$ and $p_i(E,\alpha)$, $i = H, L$ back to equations (4) and (5).

We next examine the impact of the platform's infrastructure provision and the commission rate on the subgame equilibrium, respectively. The following proposition shows how the platform's infrastructure provision affects vendors' optimal decision on individual attributes and pricing.

PROPOSITION 1 **(Impact of common attributes on individual attributes)**. *When common attributes become more desirable, in equilibrium,*

(i) firm H improves its individual attributes whereas firm L reduces its individual attributes, resulting in an enlarged quality gap. That is, $\frac{\partial q_H^*(E,\alpha)}{\partial E} > 0$ *and* $\frac{\partial q_L^*(E,\alpha)}{\partial E} < 0$.

(ii) both firms offer higher overall qualities and charge higher prices, with an increased price gap. That is, $\frac{\partial Q_H^*(E,\alpha)}{\partial E} > \frac{\partial Q_L^*(E,\alpha)}{\partial E} > 0$ *and* $\frac{\partial p_H^*(E,\alpha)}{\partial E} > \frac{\partial p_L^*(E,\alpha)}{\partial E} > 0$.

When common attributes become more desirable, one might expect vendors to slash their individual attributes so as to enjoy a cost reduction. However, Proposition 1(i) shows diverging trends in competing vendors' individual attributes as common attributes become more desirable. As illustrated in Figure 1, facing a rise in E, vendor H responds by improving q_H whereas vendor L responds by cutting q_L. This is because improved infrastructure by allowing vendor L to expand demand to the low-end market helps enhance vendor L's competition status against vendor H, suggesting a "bottom-up" impact. In our research context of digital goods, such intensified competition gives vendor H strong incentive to *increase* individual attributes and position away from vendor L. Below we detail this insight by identifying two routes through which a higher E affects competing vendors' individual attribute decisions.

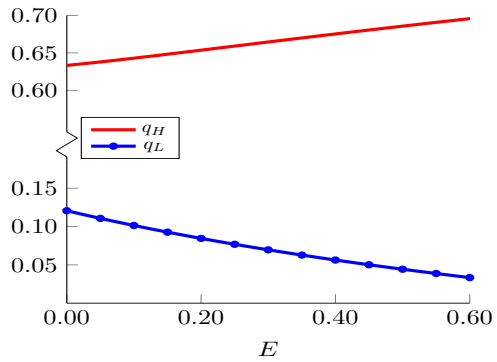


Figure 1: Impact of Common Attributes on Individual Attributes ($f = 0.4$)

First route: This route of impact comes from two effects of E.

Low-end expansion effect To see this effect, consider the case when both firms' individual attributes and prices are fixed. In this case, a larger E increases consumer utility for product H and product L by the same amount and therefore have no bearing on a consumer's preference when choosing between the two products. Meanwhile, more desirable common attributes favor the purchasing decision and convert some consumers at the low end of the market from non-buyers to buyers of product L. Similar low-end expansion effects can be observed in prior research that considers a partially covered market (e.g., Moorthy 1988, Jones and Mendelson 2011, Jing 2007).

Surplus extraction effect To see this effect, consider the case when vendors are able to adjust prices but individual attributes are fixed. In this case, a larger E motivates both vendors to raise their prices, but the *low-end expansion effect* gives vendor L more room to do so. This is because any increase in p_H will result in pure demand loss for firm H, whereas firm L can offset such demand loss by acquiring new buyers from the low-end market.[15] This additional flexibility in pricing enhances vendor L's competition status, as reflected by the reduced price gap between p_H and p_L (see equation (8)).

These two effects, when combined, form the first route of impact on vendors' individual attributes decisions. In particular, the reduced price gap suggests intensified competition, which encourages firms to position away from each other once they are able to do so (e.g., Shaked and Sutton 1982, Moorthy 1988). Importantly, the impact through this route is particularly strong for digital goods, for which the fierce price competition due to trivial marginal cost (Jones and Mendelson 2011) motivates firm L to slash q_L, but pushes firm H to increase q_H.

Second route: This route of impact comes from the following effect.

Quality substitution effect If firms can adjust individual attributes but not prices, more desirable common attributes naturally induce them to slash their individual offerings. This effect is similar to what is demonstrated in Desai (2001), Geng et al. (2022), and Lambertini and Orsini (2001) where positive externality by enhancing product valuations motivates firms to reduce quality provision. Intuitively, this effect becomes weaker under lower marginal production cost.

[15] In contrast, for horizontally differentiated firms, common attributes have symmetric impacts on firm pricing.

Whereas the three effects exist separately in other settings, our study provides an interesting context where all three are present. Because their impacts on the vendors' product decisions are not always aligned, it is unclear which effect(s) dominates at the equilibrium. Proposition 1 answers this exact question, which is new to the literature. Specifically, for digital goods with trivial marginal cost, the substitution effect is weak and so the impact of E through the second route is dominated by the first route of impact.[16] This leads to the divergence of the vendors' individual attributes as illustrated in Figure 1. In contrast, in markets where the marginal cost is substantial, the substitution effect becomes much stronger and could dominate the impact of the first route, so that firm H may reduce its individual attributes as E increases. We have conducted numerical analysis to verify this insight.

Note that more desirable common attributes increase the overall quality of both products in equilibrium with widened quality gap, which ultimately expands the price gap as indicated in Proposition 1(ii). This equilibrium result contrasts equation (8), which shows a reduced price gap if vendors do *not* adjust individual attributes in response to more desirable common attributes. This contrast highlights how vendors' strategic adjustment of individual attributes helps alleviate competition pressure.

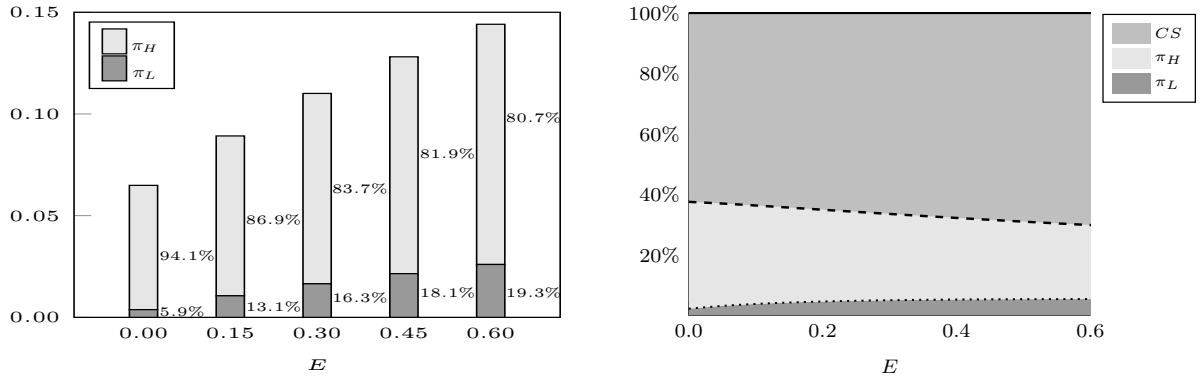
We next investigate how common attributes affect equilibrium outcome. Intuitively, when common attributes become more desirable, both firms and consumers are better off. We define profit ratio $\pi_L/(\pi_L+\pi_H)$, which indicates firm L's competition status relative to firm H, and derive consumers' total surplus as follows.

$$CS = \int_{\theta_{L0}}^{\theta_{HL}} (\theta(q_L+E)-p_L)d\theta + \int_{\theta_{HL}}^{1} (\theta(q_H+E)-p_H)d\theta. \tag{11}$$

We obtain the following corollary.

COROLLARY 1 **(Bottom-up impact of common attributes)**. *When common attributes become more desirable, in equilibrium,*

[16] In fact, for a monopoly firm with trivial marginal cost, the substitution effect has zero impact in the sense that its individual attributes do not change with more desirable attributes (detailed proof is available upon request).



(a) Industry profit allocation (b) Social welfare allocation

Figure 2: Impact of Common Attributes ($f = 0.4$, $\alpha = 0$)

(i) both firms obtain higher demands and profits, and consumers obtain greater surplus.

(ii) competition status of firm L is strengthened. That is, $\pi_L^*/(\pi_L^* + \pi_H^*)$ increases with E.

(iii) consumer surplus increases at a rate faster than the industry profit. That is, $CS^*/(\pi_L^* + \pi_H^*)$ increases with E.

Corollary 1(i) shows that, when common attributes become more desirable, in equilibrium the demand expands not only for firm L, but for firm H as well. That is, more low-end consumers are served, and some consumers will upgrade from product L to product H.

Interestingly, Corollary 1(ii) reveals asymmetric impacts of improved common attributes on firm profits—firm L experiences a higher percentage increase in profit than firm H. This suggests an improvement of firm L's competitive status, or softened market dominance of firm H. Figure 2(a) illustrates this result, i.e., $\pi_L/(\pi_H + \pi_L)$ increases with E from 5.9% (when $E = 0$) to 19.37% (when E reaches the upper bound, beyond which $q_L \leq 0$).[17] This is consistent with the "bottom-up" effect revealed by Proposition 1, which shows that common attributes more directly affect the low-quality firm than the high-quality firm, and consequently improve its profit share.

Note that improved common attributes not only improve the overall quality consumers enjoy, but also expand the market coverage, both contributing to higher consumer welfare. Corollary

[17] Note that the range of profit share (from 5.9% to 19.37%) does not depend on parameter f. See Proof of Corollary 1 for its derivation.

1(iii) further shows that, as common attributes become more appealing, consumer surplus increases faster than the industry profit. That is, the increase in industry profit does not come at a cost to consumers, who are able to retain the majority of the benefit from improved common attributes. We illustrate this result in Figure 2(b).

The platform monetizes its provision of common infrastructure by charging a transaction-based commission. Our next proposition shows that a higher commission rate has different impacts on the quality provision of the high-quality and low-quality vendors.

PROPOSITION 2 **(Top-down impact of commission rate)**. *When the commission rate α increases, in equilibrium,*

(i) both vendors' individual attributes decrease, with the high-quality vendor experiencing a larger magnitude of reduction. That is, $\frac{\partial q_H^*(E,\alpha)}{\partial \alpha} < \frac{\partial q_L^*(E,\alpha)}{\partial \alpha} < 0$.

(ii) both vendors charge a lower price with a narrowed price gap, that is, $\frac{\partial p_H^*(E,\alpha)}{\partial \alpha} < \frac{\partial p_L^*(E,\alpha)}{\partial \alpha} < 0$.

(iii) demand increases for both vendors.

Proposition 2(i) shows that a higher commission rate leads to lower quality provision of both vendors, which confirms the numerical findings for a retail platform (Zhang et al. 2019, Xiong et al. 2024). More importantly, our analysis reveals that in response to a higher commission rate, the high-quality vendor experiences a more pronounced reduction in individual attributes. As such, the commission rate exhibits a "top-down" pattern of influence, which poses an interesting contrast to the "bottom-up" influence brought by common attributes as shown in Corollary 1. Intuitively, the high-quality firm that dominates the market relinquishes a greater volume of profit than the low-quality firm when facing the shift in commission rate, and so adjusts its offering in a larger magnitude.

Conventional wisdom suggests that a higher commission rate will drive vendors to charge higher prices. Nonetheless, the higher commission rate also incentivizes vendors to cut down individual attributes, and consequently creates a downward pressure on prices. As shown in Proposition 2(ii) and (iii), the decline in prices prevails, resulting in higher demands for both vendors.

The result above highlights how the platform can use commission rate in addition to infrastructure provision to influence vendors' product strategies. We next examine the platform strategies.

4.2. Platform Strategies

Anticipating competing vendors' strategic reactions, the platform in stage 1 chooses common infrastructure E and commission rate α that maximize its profit

$$\Pi_P = \alpha(p_H D_H + p_L D_L) - \frac{\lambda E^2}{2}. \tag{12}$$

The following lemma characterizes the platform's optimal decision.

LEMMA 2 **(Optimal Platform Strategies)**. *The platform's optimal* $\{E^*, \alpha^*\}$ *satisfy*

$$E^* = \frac{1}{\lambda} \cdot \frac{(r^*-1)^2(4r^*+1)(12r^{*2}-5r^*+8)(208r^{*3}+184r^{*2}+177r^*+16)}{(1-4r^*)^2(6976r^{*6}-9168r^{*5}+13068r^{*4}-8095r^{*3}+6477r^{*2}-2532r^*+24)}, \tag{13}$$

$$\alpha^* = \frac{(4r^*+1)(12r^{*2}-5r^*+8)(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)}{r^*(6976r^{*6}-9168r^{*5}+13068r^{*4}-8095r^{*3}+6477r^{*2}-2532r^*+24)}, \tag{14}$$

where $r^* = Q_H^*/Q_L^*$ *is uniquely determined by*

$$\frac{f}{\lambda} = \frac{(4r^{*3}-23r^{*2}+12r^*-8)(3904r^{*7}-7024r^{*6}+8044r^{*5}-5265r^{*4}+5112r^{*3}-2020r^{*2}+560r^*+64)}{(1-r^*)^3(4r^*-1)(4r^*+1)(12r^{*2}-5r^*+8)(208r^{*3}+184r^{*2}+177r^*+16)}. \tag{15}$$

Lemma 2 suggests that the platform's optimal commission rate α^* is uniquely determined by the ratio f/λ, where f reflects the vendors' product development efficiency and λ reflects the platform's efficiency for providing the common infrastructure.[18] As can be seen from equation (13), λ has an additional negative impact on the optimal level of common infrastructure E^*.

Intuitively, a higher cost for building the infrastructure (i.e., a higher λ) suppresses the platform's provision of common attributes, which subsequently impacts the vendors' individual attributes and prices as identified in §4.1. It is, however, interesting to investigate how the platform chooses E and α based on f, the product development cost in the vendors' market. In practice, vendors' adoption of new product development tools can reduce f, allowing vendors to develop product features at

[18] Prior literature examines a retail platform's optimal commission rate independent of infrastructure provision (Zhang et al. 2019, Hsiao et al. 2023, Xiong et al. 2024). In our research context, competing vendors utilize the platform infrastructure to develop and distribute digital products, and the platform jointly decides the optimal infrastructure and its corresponding price, i.e., the commission rate.

a lower cost. For instance, third-party GenAI tools such as Promethean AI, Meshy.ai, Modl.ai, and Rosebud AI can help vendors on Steam.com lift productivity. Such universal improvement in vendor productivity reshapes the vertical competition, affecting the platform's strategic decisions as illustrated in the next proposition.

PROPOSITION 3 **(Infrastructure and commission rate under revenue sharing)**. *For a platform that offers common infrastructure, with a higher f for vendors to provide individual attributes,*

(i) the optimal infrastructure first increases and then decreases. That is, a threshold $\underline{f} > 0$ exists (see Appendix for its derivation) such that $\frac{\partial E^*}{\partial f} > 0$ for $f < \underline{f}$, and $\frac{\partial E^*}{\partial f} < 0$ for $f > \underline{f}$.

(ii) the optimal commission rate increases. That is, $\frac{\partial \alpha^*}{\partial f} > 0$.

With the decline of product development cost f, vendors are incentivized to provide higher individual attributes. In this case, one might expect that the platform slashes its infrastructure. Also, it is intuitive that the platform could raise its commission rate to take advantage of the efficiency gain of vendors' market. Surprisingly, Proposition 3 shows that the adjustment of the infrastructure is not monotone with the decline in vendors' development cost. Result (i) shows that when vendors' development cost f is reduced but remains relatively high, the platform invests more to improve its infrastructure, suggesting that the platform's benefit from the enhanced revenue in the vendors' market more than compensates for the higher cost of infrastructure improvement. However, if vendors' cost f is already very small and continues to decline, the platform prefers to reduce provision of infrastructure to save cost.

Meanwhile, Proposition 3(ii) shows the platform always accommodates a reduced product development cost of vendors with a lowered commission rate. This implies as vendors become more efficient in developing individual attributes, they will be taxed less by the platform, which consequently motivates vendors to provide better individual attributes as indicated by Proposition 2. As such, a positive feedback loop is instigated. We illustrate this result in Figure 3, where f declines along the horizontal axis from the right to the left.

Proposition 3 highlights the importance for the platform to consider the impact of its infrastructure and commission rate on vendors' individual attribute decisions. With the vendor technology improvement such as GenAI, the platform needs to carefully craft its response by adjusting the infrastructure and the commission rate simultaneously. This insight is elaborated in the next corollary, which shows the quality boosting impact of a strategic platform.

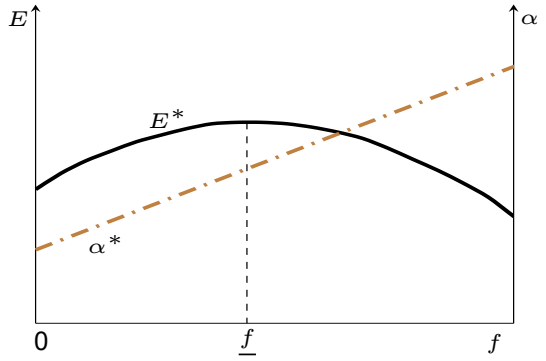


Figure 3: Platform's optimal Provision of Common Attributes and Commission Rate

Corollary 2 **(Quality boosting impact of strategic platform)**. *When vendors' development cost coefficient f declines, both q_H and q_L increase. Moreover, there exists a threshold $\underline{\underline{f}}$ (higher than $\underline{f}$), so that the platform's strategic decisions on E and α boost the increase of q_H when $f > \underline{\underline{f}}$, and boost the increase of q_L otherwise.*

Intuitively, both vendors will increase individual attributes when f becomes smaller. Corollary 2 shows that the platform's strategic response to vendors' lower development cost may bring an additional, indirect boost on their individual attributes. Interestingly, the manifestation of the boosting effect depends on the region of development cost f.

When f declines but remains above $\underline{\underline{f}}$, then $f > \underline{f}$ so that the platform responds with higher infrastructure E, but surprisingly charges a lower commission rate α (see Proposition 3). In this region, the low quality firm's incentive to raise its individual attributes due to a reduced product development cost is inhibited by the competition pressure from the high-quality firm. The increased

E and reduced α together always boost q_H, but may boost or suppress q_L at the same time (Recall Proposition 1 shows that higher E results in higher q_H and lower q_L, whereas Proposition 2 shows that lower α results in higher q_H and q_L). The additional boost in q_H allows firm H to charge a higher price while alleviating competition pressure for firm L, both effects benefiting the platform.

On the other hand, when f is below $\underline{f}$ and continues to decline, the platform reduces E and correspondingly charges a lower commission rate α. In this region, q_H is already very high, suggesting limited room for further enhancement. As a result, the platform finds it optimal to focus on boosting q_L, and the reduced E and α both serve this purpose. The further increase in q_L leads to a higher price for firm L, which makes product L less attractive and alleviates competition pressure for firm H. Both effects benefit the platform. Lastly, our analysis shows that this strategy holds optimal even when $\underline{f} < f < \underline{\underline{f}}$ (see proof of Corollary 2 for details).

Intuitively, when vendors' technology advancement brings down their development cost, the improved efficiency makes all market players better off, increasing platform payoff Π_P, vendor profits π_L and π_H, and consumer surplus CS. Our next proposition shows that the platform's strategic response to the decline of f further enhances social welfare.

PROPOSITION 4 **(Strategic response of platform)**. *When vendors' development cost coefficient f declines, the strategic response of the platform leads to higher profits of vendors and the platform, as well as a higher consumer welfare.*

Proposition 4 highlights how strategic response of the platform, when coupled with a reduced development cost in the vendors' industry, brings a win-win-win outcome to all players in the market. Whereas the platform adjusts E and α to maximize its own payoff, such adjustments help competing vendors better take advantage of the reduced f in offering individual attributes, which also benefit consumers. As a result, social welfare $SW = \Pi_P + \pi_L + \pi_H + CS$ increases. This result underlines the platform's beneficial role in coordinating the vendor market through its strategic response.

Additional analysis shows that when vendors' development cost f declines, in equilibrium the platform takes a larger share of the industry profit, that is, $\Pi_P/(\pi_L + \pi_H)$ increases (see appendix

for proof). This indicates that vendors' technology advancement strengthens the market power of the common platform they rely on. Our theoretical insight is consistent with the observation of the growing power of technology platforms, as reflected by the continuous increase of their revenue share in the digital economy (Highfill and Surfield 2022). Today, the adoption of AI tools by reducing development cost may motivate developers to offer improved individual attributes. Our study suggests that platforms are likely to free ride this leap in industry efficiency and enjoy a strengthened market status in its interaction with vendors.

Meanwhile, we find that when vendors' development cost f declines, in equilibrium although total consumer surplus increases, its share in social welfare (i.e., CS/SW) declines (see appendix for proof). This happens because consumers are passive decision makers, compared to the platform that strategically sets common infrastructure as well as commission and vendors that strategically set individual attributes and prices.

5. Model Extension

We consider extensions of the main model to demonstrate robustness and also to obtain additional insights. All proofs are relegated to Online Appendix.

5.1. Additional platform monetization source

In addition to collecting commissions, a platform often benefits from inducing a higher sales volume, $D_H + D_L$. For example, a higher transaction volume can bring better visibility, and consequently attracts more users and advertisers to the platform. We capture such benefits using a stylized expression $\rho(D_H + D_L)$, with $\rho \geq 0$ representing the marginal benefit from transaction volume on the platform. The platform's profit with mixed income sources can thus be written as $\Pi_P = \alpha(p_H D_H + p_L D_L) + \rho(D_H + D_L) - \frac{\lambda E^2}{2}$. Our next proposition compares the platform's optimal strategies with and without this additional benefit from sales volume.

PROPOSITION 5 **(Platform strategies under mixed income)**. *For a platform receiving mixed income (i.e., $\rho > 0$), it provides a higher common infrastructure E and charges a higher commission rate α than if the platform only receives commission income (i.e., $\rho = 0$).*

Proposition 5 suggests that the additional income from transaction volume motivates the platform to offer better infrastructure so that vendors can cover more of the low-end market. Also, the platform would increase the commission rate to further expand market coverage. The intuition of this result stems from Proposition 2, which shows that higher commission rate disincentivizes vendors from providing individual attributes. Prices decline as a consequence, generating a greater sales volume for the platform. Proposition 5 confirms previous insights that the platform can effectively combine the two strategic tools for better outcome.

It is worth noting that despite the benefit of improving platform revenue in the short term, the platform's strategy by raising commission rate will suppress vendor qualities. This may hurt the platform's reputation in the long run, although the increase of common infrastructure helps mitigate this effect. Prior literature has shown that managing the quality of vendors' offerings is vital for the success of a two-sided platform (Hagiu 2009, Zheng and Kaiser 2013, Huang et al. 2022). For example, the once-dominant video game platform, Atari, collapsed after poor-quality games flooded the market, which also led to the bankruptcy of over 90% of video game developers (Boudreau and Hagiu 2009). Our finding suggests the platform should carefully monitor vendor qualities (e.g., by introducing consumer reviews) when anticipating vendors adjust their individual attributes in response to the provision and pricing of the common infrastructure.

5.2. Possible full market coverage

The main model captures the realistic situation of partial market coverage, where some low-valuation consumers will not purchase from either of the two firms (e,g., Moorthy 1988, Choi and Shin 1992). We now consider a model extension where full market coverage may arise in equilibrium so that all consumers purchase a product (Gabszewicz et al. 1986). This case provides insights on the underlying drives to findings under partial market coverage.

The model setup is similar to the main model, except that consumer quality preference θ is now uniformly distributed on $[a, b]$ where $b > a > 0$. Note that ratio b/a reflects the degree of consumer heterogeneity (Wauthy 1996). When $a = 0$ and $b = 1$, $b/a = \infty$, we have the main model where the

significant consumer heterogeneity ensures that the market is partially covered. When $b \leq 2a$, firm L will exit so that the market is no longer competitive. Thus, we consider $b > 2a$ in this section.

For the subgame of vendor competition, Table 1 provides conditions for full and partial market coverage to arise in equilibrium, where $E_1 = max\{0, \frac{5b^3-38ab^2+97a^2b-136a^3}{27fab}\}$ and $E_2 = \frac{(b-2a)(a+b)(5b-7a)}{54a(b-a)f}$. As illustrated in Figure 4, there are three equilibrium regions depending on consumer heterogeneity b/a and common attributes E. In region (I) with high consumer heterogeneity, the market is partially covered in equilibrium and consumers with low quality-preferences are not served. In this region, all results in the main model continue to hold. In particular, q_H increases with E whereas q_L decreases with E. In regions (II) and (III) with low consumer heterogeneity, the market is fully covered in equilibrium. The market coverage is largely determined by consumer heterogeneity, a result consistent with the literature (Wauthy 1996). Nonetheless, desirable common attributes E expand the region of full market coverage, as illustrated in Figure 4.

	Condition	Coverage	$\frac{\partial q_H}{\partial E}$	$\frac{\partial q_L}{\partial E}$
I	$b > 4a$ & $E < E_1$	Partial	↑	↓
II	$b > 2a$ & $E_1 \leq E \leq E_2$	Full	↓	↓
III	$b > 2a$ & $E > E_2$	Full	—	↓

Table 1: Partial and Full Market Coverage

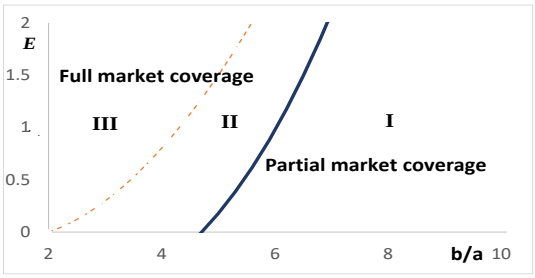


Figure 4: Market Coverage Regions

In this subgame, we find that the result of partial market coverage mostly extends to full market coverage, with two differences that highlight the critical role of the low-end expansion effect in driving the impact of common attributes in a vertically differentiated market.

First, regarding results in Proposition 1, q_H no longer increases with E, whereas other results continue to hold (i.e., the quality gap and price gap continue to increase with E; q_L still decreases with E). In particular, q_H decreases with E in region (II), whereas in region (III) common attributes have no impact on q_H, and consequently limited impact on firms' prices and demands. This is

because under full market coverage the low-end expansion effect completely vanishes. The lacking of demand expansion opportunities curtails firm L's incentive to increase its price, which weakens the surplus extraction effect. The first route of impact is thus dominated by the second route of impact (i.e., the substitution effect).

Second, regarding results in Corollary 1, our numerical analysis shows that both the profit share of firm L (π_L/π_T) and the share of consumer surplus in social welfare (CS/SW) increase with E less rapidly compared to the case of partial market coverage, and decline when E becomes sufficiently large. This is because under partial market coverage, the low-end expansion effect significantly contributes to the increase of firm L's profit and consumer surplus by converting non-buyers at the low-end market to buyers. This benefit however no longer exists under full market coverage.

Going back to Stage 1 where the platform chooses infrastructure and commission rate, we verify through numerical analysis that major results in Proposition 3 (corresponding to region I) extend to regions II and III. See Supplementary Appendix for details.

6. Conclusion

This study is inspired by the prevalence of digital platforms that provide common infrastructure to competing vendors in digital goods markets. The common infrastructure allows developers to streamline the product development process and ensures consistency in user experience. This leads to "common attributes" that are inseparable and indispensable from the competing products sold through the same platform. Meanwhile, to monetize the offering of infrastructure, platforms often charge vendors a transaction-based commission, which adds another layer of complexity in the interactions between the platform and vendors. In this research context, we examine the interactions between a platform and competing vendors by focusing on how the platform's provision and pricing of common infrastructure shape vertical competition between vendors.

Specifically, we consider a platform that offers infrastructure to a high-quality and a low-quality vendor and charges a transaction-based fee. Our investigation renders a set of interesting results. First, we find that when the platform offers more powerful infrastructure for the same commission

rate, the high-quality vendor responds with improved, rather than reduced individual attributes. Meanwhile, the low-quality firm slashes individual attributes. That is, we observe an interesting divergence in competing vendors' individual attributes. Moreover, improved common infrastructure by allowing the low-quality firm to expand sales to the low-end market exert a greater impact on the low-quality firm than on the high-quality firm, manifesting a "bottom-up" pattern of influence. As such, improved common attributes shift the market balance in favor of the low-quality firm. This insight is interesting because the high-quality firm is known to enjoy dominating power for products in digital goods market. Additionally, we show that a major portion of the benefit from more desirable common attributes is passed through to end consumers. On the other hand, when the platform charges a higher commission for the same infrastructure, both vendors expand sales by reducing individual attributes as well as product prices. This result highlights how the platform can use commission rate in addition to infrastructure provision to influence vendors' product strategies.

Second, we show that the platform will devise infrastructure strategies to support vendors' incentive to improve individual attributes when their product development cost is reduced. In particular, the platform responds to a reduced vendor development cost with more powerful infrastructure and lower commission, unless vendors' development cost is already very low. In that case, further reduction in vendors' development cost leads to less powerful infrastructure yet lower commission rate still. These findings suggest the platform might benefit from customizing infrastructure provision for markets with different product development efficiency.

Lastly, we show that the platform's strategic design of infrastructure and its pricing, when coupled with a reduced vendor development cost leads to a win-win-win outcome where all market players including the platform itself, competing vendors, and consumers are better off. This result highlights the critical role the platform plays in fueling the digital economy. Moreover, when the platform benefits from sales volume in addition to commission income, the platform would always charge a higher commission rate than if it receives commission income only.

With rapid advancements in information technology, small and innovative firms may find it beneficial to leverage the powerful infrastructure of technology platforms for new product launches.

Our research brings timely insight into quality competition in such a market environment. In practice, the platform might provide product recommendations or facilitate consumer reviews. Future research could examine how common attributes affect these strategic platform decisions. Another interesting topic is to study sequential product introduction. Wei and Nault (2008) considers how a monopolist uses product versioning to deter market entry. Further research could study the quality and price decisions of the incumbent and the entrant on a common platform.

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Online Appendix
Vertical Competition on a Common Platform

Proof of Lemma 1. Substituting equations (2)-(3) into equations (4)-(5), we can rewrite the profit functions of two vendors:

$$\pi_H = (1-\alpha)p_H\left(1-\frac{p_H-p_L}{q_H-q_L}\right)-\frac{f}{2}q_H^2,$$
$$\pi_L = (1-\alpha)p_L\left(\frac{p_H-p_L}{q_H-q_L}-\frac{p_L}{q_L+E}\right)-\frac{f}{2}q_L^2,$$

with $0 \leq \frac{p_L}{q_L+E} \leq \frac{p_H-p_L}{q_H-q_L} \leq 1$. From first-order conditions, we obtain prices as functions of individual attributes:

$$p_L(q_H,q_L) = \frac{(q_H-q_L)(q_L+E)}{4q_H-q_L+3E},$$
$$p_H(q_H,q_L) = \frac{2(q_H-q_L)(q_H+E)}{4q_H-q_L+3E},$$

with a price gap of $p_H - p_L = \frac{(2q_H-q_L+E)(q_H-q_L)}{4q_H-q_L+3E}$. Note that $q_i > 0, i = H, L$, and $E > 0$, we prove that the equilibrium prices increase with E but the price gap $p_H - p_L$ decreases with E, i.e.,

$$\frac{dp_L}{dE} = \frac{4(q_H-q_L)^2}{(4q_H-q_L+3E)^2} > 0,$$
$$\frac{dp_H}{dE} = \frac{2(q_H-q_L)^2}{(4q_H-q_L+3E)^2} > 0,$$
$$\frac{d(p_H-p_L)}{dE} = -\frac{2(q_H-q_L)^2}{(4q_H-q_L+3E)^2} < 0.$$

Plugging $p_L(q_H,q_L)$ and $p_H(q_H,q_L)$ back to the vendors' profit functions, we solve for their equilibrium individual attributes, which satisfy the first order conditions in Lemma 1. ■

Proof of Proposition 1. Plugging the price functions specified in equations (6)-(7) back into vendors' profit functions yields

$$\pi_H = \frac{4(1-\alpha)(q_H-q_L)(q_H+E)^2}{(4q_H-q_L+3E)^2}-\frac{f}{2}q_H^2, \tag{16}$$
$$\pi_L = \frac{(1-\alpha)(q_H-q_L)(q_L+E)(q_H+E)}{(4q_H-q_L+3E)^2}-\frac{f}{2}q_L^2. \tag{17}$$

Given α and E, vendors' optimal individual attributes q_H and q_L satisfy the first-order conditions:

$$(1-\alpha)(q_H+E)^2(4q_H-7q_L-3E)-fq_L(4q_H-q_L+3E)^3=0,$$

$$4(1-\alpha)(q_H+E)\left(2(q_L+E)^2-3(q_L+E)(q_H+E)+4(q_H+E)^2\right)-fq_H(4q_H-q_L+3E)^3=0.$$

Recall $Q_i=q_i+E$, $i=H,L$ and define $r=Q_H/Q_L>1$. We rewrite the first-order conditions as:

$$(1-\alpha)r^2(4r-7)-f(Q_L-E)(4r-1)^3=0, \tag{18}$$

$$4(1-\alpha)r(2-3r+4r^2)-f(rQ_L-E)(4r-1)^3=0. \tag{19}$$

Combining equations (18)-(19) yields firm L's overall quality $Q_L^*=\frac{r^*(12r^{*2}-5r^*+8)(1-\alpha)}{f(r^*-1)(4r^*-1)^3}$, where $r^*(E,\alpha)$ satisfies

$$(1-\alpha)r^*(8-12r^*+23r^{*2}-4r^{*3})=f(r^*-1)(4r^*-1)^3E. \tag{20}$$

This implies $Q_H^*=r^*Q_L^*=\frac{r^{*2}(12r^{*2}-5r^*+8)(1-\alpha)}{f(r^*-1)(4r^*-1)^3}$. Rearranging terms of equation (20) yields

$$\frac{r^*(8-12r^*+23r^{*2}-4r^{*3})}{(r^*-1)(4r^*-1)^3}=\frac{fE}{1-\alpha}.$$

Taking derivative with respect to r^* on its left side yields $\frac{64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8}{(4r^*-1)^4(r^*-1)^2}$, which is positive on $r^*>1$. Thus, given any E and α, there exists a unique $r^*(E,\alpha)$ that satisfies this equation. From equation (20), we can write $E=\frac{(1-\alpha)r^*(8-12r^*+23r^{*2}-4r^{*3})}{f(r^*-1)(4r^*-1)^3}$, which can be used to derive the equilibrium individual attributes

$$q_L^*(E,\alpha)=Q_L^*-E=\frac{r^{*2}(4r^*-7)(1-\alpha)}{f(4r^*-1)^3}, \tag{21}$$

$$q_H^*(E,\alpha)=Q_H^*-E=\frac{4r^*(4r^{*2}-3r^*+2)(1-\alpha)}{f(4r^*-1)^3}. \tag{22}$$

Below we show that r^* can only take values in an interval $[\underline{r},\bar{r}]$. First, for $q_L^*\geq 0$ to hold, we must have $r^*\geq\underline{r}=\frac{7}{4}$. Second, recall that $r^*>1$ because $Q_H>Q_L$. Also, $f>0$ and $E\geq 0$. Equation (20) thus suggests $8-12r^*+23r^{*2}-4r^{*3}\geq 0$. It is easy to verify that there exists an upper bound $\bar{r}=\frac{1}{12}\left(\sqrt[3]{24\sqrt{39279}+8927}+\sqrt[3]{8927-24\sqrt{39279}}+23\right)\approx 5.25123$ such that the expression stays negative for any $r^*>\bar{r}$. Note that $\underline{r}$ and $\bar{r}$ do *not* depend on f.

Next, we show how q_L^*, q_H^* and $q_H^* - q_L^*$ vary with E. Taking derivative of both sides of equation (20) with respect to E and rearranging terms, we obtain

$$\frac{\partial r^*(E,\alpha)}{\partial E} = -\frac{f(4r^*-1)^4(r^*-1)^2}{(-8-40r^*+63r^{*2}-34r^{*3}+64r^{*4})(1-\alpha)} < 0 \tag{23}$$

for $\underline{r} < r^* < \bar{r}$. The one-to-one negative correspondence between r^* and E indicates that E takes value on $[0, \bar{E}_r]$. For $E > \bar{E}_r$, we have $r^* < \underline{r}$ and $q_L^* < 0$. We prove for $\underline{r} < r^* < \bar{r}$

$$\begin{aligned}
&\frac{\partial q_L^*}{\partial E} = \frac{2r^*(7+8r^*)(1-\alpha)}{f(4r^*-1)^4}\frac{\partial r^*}{\partial E} < 0, \\
&\frac{\partial q_H^*}{\partial E} = -\frac{8(1+5r^*)(1-\alpha)}{f(4r^*-1)^4}\frac{\partial r^*}{\partial E} > 0, \text{ and so} \\
&\frac{\partial (q_H^* - q_H^*)}{\partial E} > 0.
\end{aligned}$$

We also verify that the overall qualities of both firms increase with E for $\underline{r} \le r^* \le \bar{r}$:

$$\begin{aligned}
\frac{\partial Q_L^*}{\partial E} &= \frac{\partial (q_L^* + E)}{\partial E} = \frac{(8+54r^*-75r^{*2}+16r^{*3}-48r^{*4})(1-\alpha)}{f(4r^*-1)^4(r^*-1)^2}\frac{\partial r^*}{\partial E} > 0, \\
\frac{\partial Q_H^*}{\partial E} &= \frac{\partial (q_H^* + E)}{\partial E} = \frac{r^*(16+9r^*-6r^{*2}-64r^{*3})(1-\alpha)}{f(4r^*-1)^4(r^*-1)^2}\frac{\partial r^*}{\partial E} > 0.
\end{aligned}$$

Substituting q_L^* and q_H^* as specified in equations (21)-(22) back into price functions in equations (6)-(7) yields equilibrium prices in the subgame:

$$p_H^* = \frac{2r^{*2}(12r^{*2}-5r^*+8)(1-\alpha)}{f(4r^*-1)^4}, \tag{24}$$

$$p_L^* = \frac{r^*(12r^{*2}-5r^*+8)(1-\alpha)}{f(4r^*-1)^4}, \text{with} \tag{25}$$

$$p_H^* - p_L^* = \frac{r^*(24r^{*3}-22r^{*2}+21r^*-8)(1-\alpha)}{f(4r^*-1)^4}.$$

We prove that for $\underline{r} \le r^* \le \bar{r}$,

$$\begin{aligned}
\frac{\partial p_L^*}{\partial E} &= -\frac{2(24r^{*3}-2r^{*2}+43r^*+4)(1-\alpha)}{f(4r^*-1)^5}\frac{\partial r^*}{\partial E} > 0, \\
\frac{\partial p_H^*}{\partial E} &= -\frac{2r^*(28r^{*2}+49r^*+16)(1-\alpha)}{f(4r^*-1)^5}\frac{\partial r^*}{\partial E} > 0, \text{ and} \\
\frac{\partial (p_H^* - p_L^*)}{\partial E} &= -\frac{2(r^{*3}+51r^{*2}-27r^*-4)(1-\alpha)}{f(4r^*-1)^5}\frac{\partial r^*}{\partial E} > 0.
\end{aligned}$$

■

Proof of Corollary 1. We obtain vendors' equilibrium profits by substituting q_L^* and q_H^* as specified in equations (21)-(22) back into profit functions as specified in equations (16)-(17):

$$\pi_L^* = \frac{r^{*2}\left(-16r^{*4}+152r^{*3}-113r^{*2}+74r^*-16\right)(1-\alpha)^2}{2f(4r^*-1)^6}, \tag{26}$$

$$\pi_H^* = \frac{4r^{*2}\left(16r^{*4}+16r^{*3}-13r^{*2}+16r^*-8\right)(1-\alpha)^2}{f(4r^*-1)^6}. \tag{27}$$

We verify for $\underline{r} \le r^* \le \bar{r}$, $0<\alpha<1$ and $\frac{\partial r^*}{\partial E}<0$,

$$\frac{\partial \pi_L^*}{\partial E} = -\frac{r^*\left(256r^{*4}-72r^{*3}+218r^{*2}-17r^*-16\right)(1-\alpha)^2}{f(4r^*-1)^7}\frac{\partial r^*}{\partial E} > 0,$$

$$\frac{\partial \pi_H^*}{\partial E} = -\frac{16r^*\left(40r^{*4}-6r^{*3}+35r^{*2}-20r^*-4\right)(1-\alpha)^2}{f(4r^*-1)^7}\frac{\partial r^*}{\partial E} > 0.$$

Firm L's profit share is $\frac{\pi_L^*}{\pi_H^*+\pi_L^*} = \frac{16r^{*4}-152r^{*3}+113r^{*2}-74r^*+16}{-112r^{*4}-280r^{*3}+217r^{*2}-202r^*+80}$. For $\underline{r} \le r^* \le \bar{r}$, we have

$$\frac{\partial(\pi_L^*/(\pi_L^*+\pi_H^*))}{\partial r^*} = -\frac{48\left(448r^{*6}-672r^{*5}+748r^{*4}-672r^{*3}+621r^{*2}-232r^*+56\right)}{\left(112r^{*4}+280r^{*3}-217r^{*2}+202r^*-80\right)^2} < 0.$$

Because $\frac{\partial r^*}{\partial E}<0$, we thus must have $\frac{\partial(\pi_L^*/(\pi_L^*+\pi_H^*))}{\partial E} = \frac{\partial(\pi_L^*/(\pi_L^*+\pi_H^*))}{\partial r^*}\frac{\partial r^*}{\partial E} > 0$. Recall that the upper and lower bounds of r^* do not depend on f. Therefore, $\pi_L^*/(\pi_L^*+\pi_H^*)$ increases from about 5.9% to 19.37% for $\underline{r} \le r^* \le \bar{r}$ (i.e., $0 \le E \le \bar{E}_r$). ■

Proof of Proposition 2. (i) Taking derivative of both sides of equation (20) with respect to α and rearranging terms, we obtain for $\underline{r} \le r^* \le \bar{r}$

$$\frac{\partial r^*(E,\alpha)}{\partial \alpha} = -\frac{r^{*2}(8-12r^*+23r^{*2}-4r^{*3})^2}{f(4r^*-1)^2(-8-40r^*+63r^{*2}-34r^{*3}+64r^{*4})E} < 0. \tag{28}$$

Thus, given (E,α) we prove for $\underline{r} \le r^* \le \bar{r}$,

$$\frac{\partial q_L^*}{\partial \alpha} = \frac{(48r^{*4}-40r^{*3}+67r^{*2}-176r^*+56)E}{(4r^{*3}+23r^{*2}-12r^*+8)^2}\frac{\partial r^*}{\partial \alpha} < 0,$$

$$\frac{\partial q_H^*}{\partial \alpha} = \frac{4(64r^{*4}-56r^{*3}+41r^{*2}-20r^*+16)E}{(4r^{*3}+23r^{*2}-12r^*+8)^2}\frac{\partial r^*}{\partial \alpha} < 0, \text{ and}$$

$$\frac{\partial(q_H^*-q_L^*)}{\partial \alpha} = \frac{(208r^{*4}-184r^{*3}+97r^{*2}+96r^*+8)E}{(4r^{*3}+23r^{*2}-12r^*+8)^2}\frac{\partial r^*}{\partial \alpha} < 0.$$

(ii) We next show that the equilibrium prices and the price gap decrease with α. By equations (24)-(25), we prove for $\underline{r} \le r^* \le \bar{r}$, $E>0$ and $\frac{\partial r^*}{\partial \alpha}<0$,

$$\frac{\partial p_L^*}{\partial \alpha} = \frac{(192r^{*6}-544r^{*5}+1404r^{*4}-1952r^{*3}+2191r^{*2}-864r^*+248)E}{(4r^*-1)^2(-4r^{*3}+23r^{*2}-12r^*+8)^2}\frac{\partial r^*}{\partial \alpha} < 0,$$

$$\frac{\partial p_H^*}{\partial \alpha} = \frac{2(880r^{*6} - 1288r^{*5} + 1159r^{*4} - 344r^{*3} + 412r^{*2} - 208r^* + 64)E}{(4r^* - 1)^2(-4r^{*3} + 23r^{*2} - 12r^* + 8)^2} \frac{\partial r^*}{\partial \alpha} < 0, \text{ and}$$
$$\frac{\partial (p_H^* - p_L^*)}{\partial \alpha} = \frac{(1568r^{*6} - 2032r^{*5} + 914r^{*4} + 1264r^{*3} - 1367r^{*2} + 448r^* - 120)E}{(4r^* - 1)^2(-4r^{*3} + 23r^{*2} - 12r^* + 8)^2} \frac{\partial r^*}{\partial \alpha} < 0.$$

(iii) Substituting equations (21)-(22) and (24)-(25) into equations (2)-(3), we have $\theta_{L0}^* = \frac{r^*-1}{4r^*-1}$ and $\theta_{HL}^* = \frac{2r^*-1}{4r^*-1}$. Thus, $D_H^* = \frac{2r^*}{4r^*-1}$ and $D_L^* = \frac{r^*}{4r^*-1}$, so that with $\frac{\partial r^*}{\partial \alpha} < 0$, we must have $\frac{\partial D_L^*}{\partial \alpha} = \frac{1}{(4r^*-1)^2} \frac{\partial r^*}{\partial \alpha} < 0$ and $\frac{\partial D_H^*}{\partial \alpha} = \frac{2}{(4r^*-1)^2} \frac{\partial r^*}{\partial \alpha} < 0$.

We further show that a simultaneous increase in both E and α results in lower values of $r^*(E, \alpha)$, a result we will later use to prove proposition 5. To see that, we rewrite equation (20) as

$$\frac{1-\alpha}{E} = \frac{f(r^* - 1)(4r^* - 1)^3}{r^*(8 - 12r^* + 23r^{*2} - 4r^{*3})}.$$

Given f, $r^*(E, \alpha)$ is uniquely determined by the ratio $\frac{1-\alpha}{E}$. We have already proved $\frac{\partial r^*(E,\alpha)}{\partial E} < 0$ in equation (23) and $\frac{\partial r^*(E,\alpha)}{\partial \alpha} < 0$ in equation (28). Thus, a simultaneous increase in values of E and α, which lowers the value of $\frac{1-\alpha}{E}$, will result in lower $r^*(E, \alpha)$. ■

Proof of Lemma 2. Plugging vendors' equilibrium individual attributes q_L^* and q_H^* as specified in equations (21)-(22) and equilibrium prices p_L^* and p_H^* as specified in equations (24)-(25) to the platform's payoff function as specified in equation (12), we obtain:

$$\Pi_P = \frac{\alpha(1-\alpha)r^{*2}(1 + 4r^*)(8 - 5r^* + 12r^{*2})}{f(4r^* - 1)^5} - \frac{\lambda(1-\alpha)^2 r^{*2}(8 - 12r^* + 23r^{*2} - 4r^{*3})^2}{2f^2(r^* - 1)^2(4r^* - 1)^6}, \tag{29}$$

where $r^*(E, \alpha)$ satisfies equation (20).

Taking derivative of Π_P in equation (29) with respect to E and setting it to zero yield

$$\frac{\partial \Pi_P}{\partial E} = \frac{\partial \Pi_P}{\partial r^*} \frac{\partial r^*}{\partial E} = -\frac{\alpha(1-\alpha)r^* (208r^{*3} + 184r^{*2} + 177r^* + 16)}{f(4r^* - 1)^6} \frac{\partial r^*}{\partial E}$$
$$- \frac{(1-\alpha)^2 r^*(4r^{*3} - 23r^{*2} + 12r^* - 8)(64r^{*4} - 34r^{*3} + 63r^{*2} - 40r^* - 8)}{f^2(r^* - 1)^3(4r^* - 1)^7} \frac{\partial r^*}{\partial E} = 0.$$

Substituting $\frac{\partial r^*}{\partial E}$ as specified in equation (23) into the above equation yields

$$\frac{\partial \Pi_P}{\partial E} = \frac{r^*(1-\alpha)\lambda(4r^{*3} - 23r^{*2} + 12r^* - 8)}{f(r^* - 1)(4r^* - 1)^3} + \frac{\alpha r^*(r^* - 1)^2(208r^{*3} + 184r^{*2} + 177r^* + 16)}{(4r^* - 1)^2(64r^{*4} - 34r^{*3} + 63r^{*2} - 40r^* - 8)} = 0. \tag{30}$$

Similarly, substituting $\frac{\partial r^*}{\partial \alpha}$ as specified in equation (28) into the first-order condition $\frac{\partial \Pi_p}{\partial \alpha} = 0$ yields

$$\frac{\partial \Pi_P}{\partial \alpha} = \frac{(1-\alpha)r^{*2}(4r^*+1)\left(12r^{*2}-5r^*+8\right)\left(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8\right)-\alpha r^{*2}B}{f(4r^*-1)^5(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)} = 0, \quad (31)$$

where $B = 3904r^{*7} - 7024r^{*6} + 8044r^{*5} - 5265r^{*4} + 5112r^{*3} - 2020r^{*2} + 560r^* + 64$.

We next show the Hessian matrix $\begin{pmatrix} \frac{\partial^2 \Pi_P}{\partial \alpha \partial E} & \frac{\partial^2 \Pi_P}{\partial \alpha^2} \\ \frac{\partial^2 \Pi_P}{\partial E^2} & \frac{\partial^2 \Pi_P}{\partial E \partial \alpha} \end{pmatrix}$ is negative definite. The second-order derivatives are given as follows:

$$\begin{aligned}
\frac{\partial^2 \Pi_P}{\partial \alpha^2} &= -\frac{2r^{*2}((64r^4-34r^3+63r^2-40r-8)^2B - r^*\alpha g_1(r^*))}{f(1-\alpha)(4r^*-1)^5(64r^4-34r^3+63r^2-40r-8)^3} < 0, \\
\frac{\partial^2 \Pi_P}{\partial E^2} &= -\frac{2f\alpha(r^*-1)^3 g_2(r^*)}{(1-\alpha)(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)^3} - \lambda < 0, \\
\frac{\partial^2 \Pi_P}{\partial \alpha \partial E} &= \frac{r^*(r^*-1)^2(2(4r^{*3}-23r^{*2}+12r^*-8)g_3(r^*)+3(1-\alpha)r^* g_4(r^*))}{(1-\alpha)(4r^*-1)^2(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)^3},
\end{aligned}$$

where $g_1(r^*) > 0$, $g_2(r^*) > 0$, $g_3(r^*) > 0$ and $g_4(r^*) > 0$ for $\underline{r} \le r^* \le \bar{r}$, so that $\frac{\partial^2 \Pi_P}{\partial \alpha^2}\frac{\partial^2 \Pi_P}{\partial E^2} > (\frac{\partial^2 \Pi_P}{\partial \alpha \partial E})^2$ at (E^*, α^*). In above, the expression of $g_1(r^*)$ is cumbersome and can be found in supplementary appendix (similarly, we omit the expressions for $g_2(r^*)$ through $g_{29}(r^*)$).

Thus, there exist unique $\{E^*, \alpha^*\}$ that satisfy equations (30) and (31). Combining equations (20), (30) and (31), we have

$$\begin{aligned}
E^* &= \frac{(r^*-1)^2(4r^*+1)(12r^{*2}-5r^*+8)(208r^{*3}+184r^{*2}+177r^*+16)}{\lambda(4r^*-1)^2 A}, \\
\alpha^* &= \frac{(4r^*+1)(12r^{*2}-5r^*+8)(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)}{r^* A},
\end{aligned}$$

where $A = 6976r^{*6} - 9168r^{*5} + 13068r^{*4} - 8095r^{*3} + 6477r^{*2} - 2532r^* + 24$ and $r^*(E^*, \alpha^*)$ is obtained by plugging E^* and α^* back into $r^*(E, \alpha)$ as specified in equation (20) so that it uniquely satisfies

$$\frac{f}{\lambda} = \frac{(4r^{*3}-23r^{*2}+12r^*-8)B}{(1-r^*)^3(4r^*-1)(4r^*+1)(12r^{*2}-5r^*+8)(208r^{*3}+184r^{*2}+177r^*+16)}.$$

That is, given λ and f, there exists a unique $r^*(E^*, \alpha^*)$ that satisfies this equation.

The proof above suggests that given E, there exists a unique $\alpha^*(E)$ that maximizes Π_P, a result we will later use to prove proposition 5. ■

Proof of Proposition 3. Equation (15) gives $r^*(E^*, \alpha^*)$ under the platform's optimal strategies. Taking derivative with respect to f on both side of this equation, we prove for $\underline{r} \le r^*(E^*, \alpha^*) \le \bar{r}$,

$$\frac{\partial r^*}{\partial f} = -\frac{(4r^*-1)^2(4r^*+1)^2(r^*-1)^4(12r^{*2}-5r^*+8)^2(208r^{*3}+184r^{*2}+177r^*+16)^2}{\lambda g_5(r^*)} < 0, \quad (32)$$

where $g_5(r^*) > 0$ on $[\underline{r}, \bar{r}]$ and its expression can be found in supplementary appendix.

(i) The partial derivative of E^* as specified in equation (20) with respect to f is:

$$\frac{\partial E^*}{\partial f} = \frac{\partial E^*}{\partial r^*}\frac{\partial r^*}{\partial f} = \frac{(r^*-1)g_6(r^*)}{\lambda(4r^*-1)^3A^2}\frac{\partial r^*}{\partial f}$$

where the expression of $g_6(r^*)$ can be found in supplementary appendix. We prove that $g_6(r^*) > 0$ when $\underline{r} \le r^* < \tilde{r} \approx 5.22772$ and $g_6(r^*) < 0$ when $\tilde{r} < r^* \le \bar{r}$. Recall that $\lambda > 0$, $\underline{r} \le r^* \le \bar{r}$ and $\frac{\partial r^*}{\partial f} < 0$, we know $\frac{\partial E^*}{\partial f} < 0$ when $\underline{r} \le r^* \le \tilde{r}$ and $\frac{\partial E^*}{\partial f} > 0$ when $\tilde{r} \le r^* \le \bar{r}$. At $\tilde{r} \approx 5.22772$, by equation (15) the corresponding $\underline{f} = 0.00195208\lambda$. Because $\frac{\partial r^*}{\partial f} < 0$, the one-to-one negative correspondence between r^* and f indicates that E^* increases with f when $f \le \underline{f}$ and decreases when $f \ge \underline{f}$.

(ii) We next prove that the optimal commission rate α^* increases with f. The partial derivative of α^* with respect to f is:

$$\frac{\partial \alpha^*}{\partial f} = \frac{\partial \alpha^*}{\partial r^*}\frac{\partial r^*}{\partial f} = -\frac{g_7(r^*)}{r^{*2}A^2}\frac{\partial r^*}{\partial f} < 0, \quad (33)$$

because $\frac{\partial r^*}{\partial f} < 0$ and $g_7(r^*) > 0$ on $[\underline{r}, \bar{r}]$ (see supplementary appendix for $g_7(r^*)$). ■

Proof of Corollary 2. For exogenously given E and α, let $\mathring{r} = r^*(E, \alpha)$ as given by equation (20) with the corresponding $\mathring{q}_i(E, \alpha), i = H, L$. From equations (21) and (22) we obtain:

$$\frac{\partial \mathring{q}_L(E,\alpha)}{\partial f} = -\frac{\mathring{r}(1-\alpha)(4\mathring{r}-7)}{f^2(4\mathring{r}-1)^3} + \frac{2\mathring{r}(1-\alpha)(7+8\mathring{r})}{f(4\mathring{r}-1)^4}\frac{\partial \mathring{r}}{\partial f},$$
$$\frac{\partial \mathring{q}_H(E,\alpha)}{\partial f} = -\frac{4\mathring{r}(1-\alpha)(4\mathring{r}^2-3\mathring{r}+2)}{f^2(4\mathring{r}-1)^3} - \frac{8(1-\alpha)(1+5\mathring{r})}{f(4\mathring{r}-1)^4}\frac{\partial \mathring{r}}{\partial f}.$$

We prove that for $\underline{r} \le \mathring{r} \le \bar{r}$

$$\frac{\partial \mathring{r}}{\partial f} = \frac{\mathring{r}(\mathring{r}-1)(4\mathring{r}-1)(4\mathring{r}^3-23\mathring{r}^2+12\mathring{r}-8)}{f(64\mathring{r}^4-34\mathring{r}^3+63\mathring{r}^2-40\mathring{r}-8)} < 0. \quad (34)$$

When α^* and E^* are endogenous, taking the derivative of $q_i^*, i=H,L$ with respect to f yields

$$\frac{\partial q_L^*(E^*,\alpha^*)}{\partial f}=-\frac{r^{*2}(1-\alpha^*)(4r^*-7)}{f^2(4r^*-1)^3}+\frac{2r^*(1-\alpha^*)(7+8r^*)}{f^(4r^*-1)^4}\frac{\partial r^*}{\partial f}-\frac{(4r^*-7)r^{*2}}{f^2(4r^*-1)^3}\frac{\partial \alpha^*}{\partial f},$$
$$\frac{\partial q_H^*(E^*,\alpha^*)}{\partial f}=-\frac{4r^*(1-\alpha^*)(4r^{*2}-3r^*+2)}{f^2(4r^*-1)^3}-\frac{8(1-\alpha^*)(1+5r^*)}{f(4r^*-1)^4}\frac{\partial r^*}{\partial f}-\frac{4r^*(4r^{*2}-3r+2)}{f^2(4r^*-1)^3}\frac{\partial \alpha^*}{\partial f},$$

where $r^*(E^*,\alpha^*)$ is specified in equation (15), with $\frac{\partial r^*}{\partial f}$ and $\frac{\partial \alpha^*}{\partial f}$ specified in equations (32) and (33), respectively.

Note that at E^* and α^*, $\mathring{r}=r^*(E^*,\alpha^*)$. We thus have

$$\frac{\partial \mathring{q}_L}{\partial f}-\frac{\partial q_L^*}{\partial f}=\frac{2r^*(1-\alpha^*)(7+8r^*)}{f(4r^*-1)^4}\left(\frac{\partial \mathring{r}}{\partial f}-\frac{\partial r^*}{\partial f}\right)+\frac{(4r^*-7)r^{*2}}{f^2(4r^*-1)^3}\frac{\partial \alpha^*}{\partial f},$$
$$\frac{\partial \mathring{q}_H}{\partial f}-\frac{\partial q_H^*}{\partial f}=-\frac{8f(1-\alpha^*)(1+5r^*)}{f^2(4r^*-1)^4}\left(\frac{\partial \mathring{r}}{\partial f}-\frac{\partial r^*}{\partial f}\right)+\frac{4r^*(4r^{*2}-3r+2)}{f^2(4r^*-1)^3}\frac{\partial \alpha^*}{\partial f}.$$

Recall that $\frac{\partial \alpha^*}{\partial f}>0$, $\underline{r}\le r^*\le \bar{r}$, it follows that $\frac{\partial \mathring{q}_L^*}{\partial f}>\frac{\partial q_L^*}{\partial f}$ holds if $\frac{\partial \mathring{r}}{\partial f}>\frac{\partial r^*}{\partial f}$ and $\frac{\partial \mathring{q}_H^*}{\partial f}>\frac{\partial q_H^*}{\partial f}$ holds if $\frac{\partial \mathring{r}}{\partial f}<\frac{\partial r^*}{\partial f}$.

Substituting α^* and E^* into equation (34) and comparing it with equation (32), we have $\frac{\partial \mathring{r}}{\partial f}>\frac{\partial r^*}{\partial f}$ when $\bar{r}\ge r^*\ge \dot{r}\approx 2.91315$. Because $\frac{\partial r^*}{\partial f}<0$, the one-to-one correspondence between r^* and f indicates that $\frac{\partial \mathring{r}}{\partial f}>\frac{\partial r^*}{\partial f}$ if and only if $f<\underline{\underline{f}}=0.434584\lambda$. Clearly, $\underline{\underline{f}}$ is higher than $\underline{f}=0.00195208\lambda$ (see proof of Proposition 3). In summary, $0>\frac{\partial \mathring{q}_L^*}{\partial f}>\frac{\partial q_L^*}{\partial f}$ when $f<\underline{\underline{f}}$, and $0>\frac{\partial \mathring{q}_H^*}{\partial f}>\frac{\partial q_H^*}{\partial f}$ when $f>\underline{\underline{f}}$.■

Proof of Proposition 4. (i) We first show that given exogenous E and α, a smaller f increases vendors' payoffs and consumer surplus in the subgame equilibrium. Define $\mathring{r}=r^*(E,\alpha)$ given by equation (20). The vendors' subgame equilibrium payoffs can be obtained by replacing r^* in equations (26)-(27) with $\mathring{r}$:

$$\mathring{\pi}_L(E,\alpha)=\frac{\mathring{r}^2\left(-16\mathring{r}^4+152\mathring{r}^3-113\mathring{r}^2+74\mathring{r}-16\right)(1-\alpha)^2}{2f(4\mathring{r}-1)^6},$$
$$\mathring{\pi}_H(E,\alpha)=\frac{4\mathring{r}^2\left(16\mathring{r}^4+16\mathring{r}^3-13\mathring{r}^2+16\mathring{r}-8\right)(1-\alpha)^2}{f(4\mathring{r}-1)^6}.$$

We also derive

$$\mathring{CS}(E,\alpha)=\frac{(1-\alpha)\mathring{r}^3(4\mathring{r}+5)(12\mathring{r}^2-5\mathring{r}+8)}{2f(\mathring{r}-1)(4\mathring{r}-1)^5}, \tag{35}$$
$$\mathring{SW}(E,\alpha)=\frac{\mathring{r}^2(1-\alpha)g_4(\mathring{r})}{2f^2(4\mathring{r}-1)^6(\mathring{r}-1)^2}. \tag{36}$$

We take derivatives of the above equations with respect to f and prove for $\underline{r} \leq r \leq \bar{r}$,

$$\frac{\partial \mathring{\pi}_L}{\partial f} = -\frac{(1-\alpha)^2 \mathring{r}^2 (1024\mathring{r}^8 - 4128\mathring{r}^7 + 10144\mathring{r}^6 - 8394\mathring{r}^5 + 6415\mathring{r}^4 - 1688\mathring{r}^3 - 16\mathring{r}^2 + 416\mathring{r} - 128)}{2f^2(4\mathring{r}-1)^6(64\mathring{r}^4 - 34\mathring{r}^3 + 63\mathring{r}^2 - 40\mathring{r} - 8)} < 0, \quad (37)$$

$$\frac{\partial \mathring{\pi}_H}{\partial f} = -\frac{4(1-\alpha)^2 \mathring{r}^2 (1664\mathring{r}^8 - 3936\mathring{r}^7 + 6440\mathring{r}^6 - 6306\mathring{r}^5 + 6113\mathring{r}^4 - 3688\mathring{r}^3 + 1120\mathring{r}^2 - 128\mathring{r} - 64)}{f^2(4\mathring{r}-1)^6 (64\mathring{r}^4 - 34\mathring{r}^3 + 63\mathring{r}^2 - 40\mathring{r} - 8)} < 0, \quad (38)$$

$$\frac{\partial \mathring{CS}}{\partial f} = -\frac{(1-\alpha)\mathring{r}^3 (5440\mathring{r}^6 - 7536\mathring{r}^5 + 4428\mathring{r}^4 - 5601\mathring{r}^3 + 1032\mathring{r}^2 + 312\mathring{r} - 640)}{2f^2(4\mathring{r}-1)^5(64\mathring{r}^4 - 34\mathring{r}^3 + 63\mathring{r}^2 - 40\mathring{r} - 8)} < 0, \quad (39)$$

for $\underline{r} \leq r^* \leq \bar{r}$.

(ii) Next we examine how f affects the equilibrium payoffs under the platform's endogenous E^* and α^*. In this case, we substitute E^*, α^* and r^* as specified in equations (13)-(15) into equations (26)-(27) and equation (35) yields

$$\pi_L^* = \frac{(-16r^{*4} + 152r^{*3} + 113r^{*2} - 74r^* + 16)B^2}{2f(4r^*-1)^6 A^2},$$
$$\pi_H^* = \frac{4(16*r^4 + 16r^{*3} - 13r^{*2} + 16r^* - 8)B^2}{f(4r^*-1)^6 A^2},$$
$$CS^* = \frac{r^{*2}(4r^*+5)(12r^{*2} - 5r^* + 8)B}{2f(r^*-1)(4r^*-1)^5 A}.$$

We prove that for $\underline{r} \leq r^* \leq \bar{r}$,

$$\frac{\partial \pi_L^*}{\partial f} = -\frac{g_9(r^*)B^2}{2f^2(4r^*-1)^6 A^3 g_8(r^*)} < 0, \quad (40)$$

$$\frac{\partial \pi_H^*}{\partial f} = -\frac{4g_{10}(r^*)B^2}{f^2(4r^*-1)^6 A^3 g_8(r^*)} < 0, \quad (41)$$

$$\frac{\partial CS^*}{\partial f} = -\frac{r^*(8 - 5r^* + 12r^{*2})g_{11}(r^*)B^2}{2f^2(r^*-1)(4r^*-1)^5 A^2 g_8(r^*)} < 0, \quad (42)$$

because $g_8(r^*) > 0$, $g_9(r^*) > 0$, $g_{10}(r^*) > 0$ and $g_{11}(r^*) > 0$ on $\underline{r} \leq r^* \leq \bar{r}$ (see supplementary appendix for their expressions).

Intuitively, the platform's strategic response to the decline of f improves its own profit. We find that such strategic response improves vendors and consumer surplus as well. Substituting α^*, E^* and f specified in equations (11)-(13) into equations (37)-(39) and comparing them with equations (40)-(42), we have $\frac{\partial \mathring{\pi}_i}{\partial f} < \frac{\partial \pi_i^*}{\partial f}, i = H, L$ and $\frac{\partial \mathring{CS}}{\partial f} < \frac{\partial CS^*}{\partial f}$ for $\underline{r} \leq r^* \leq \bar{r}$ and $f > 0$. ■

Proof of results about $\Pi_P/(\pi_L+\pi_H)$ and CS/SW. We next show that as f increases, the profit ratio $\Pi_P^*/(\pi_L^*+\pi_H^*)$ decreases whereas the consumer surplus share CS/SW increases. Under endogenous E^* and α^*, the platform's profit ratio against the vendors' profit in equilibrium is

$$\frac{\Pi_P^*}{\pi_L^*+\pi_H^*}=\frac{r^*(4r^*-1)(4r^*+1)(12r^{*2}-5r^*+8)A}{(112r^{*4}+280r^{*3}-217r^{*2}+202r^*-80)B}.$$

We prove that for $\underline{r}\le r^*\le \bar{r}$, $\frac{\partial(\Pi_P^*/(\pi_L^*+\pi_H^*))}{\partial f}=\frac{g_{12}(r^*)}{(112r^{*4}+280r^{*3}-217r^{*2}+202r^*-80)^2B^2}\frac{\partial r^*}{\partial f}<0$, because $g_{12}(r^*)>0$ and $\frac{\partial r^*}{\partial f}<0$. Thus, $\Pi_P^*/(\pi_L^*+\pi_H^*)$ decreases with f.

The consumer surplus share can be written as

$$\frac{CS}{SW}=\frac{r^{*2}(4r^*-1)(4r^*+5)(12r^{*2}-5r^*+8)A}{g_{13}(r^*)}.$$

We prove that $\frac{\partial(CS/SW)}{\partial f}=-\frac{r^*g_{14}(r^*)}{g_{13}^2(r^*)}\frac{\partial r^*}{\partial f}>0$ because $g_{14}(r^*)>0$ on $\underline{r}\le r^*\le\bar{r}$ (see supplementary appendix for expressions of $g_{13}(r^*)$ and $g_{14}(r^*)$). ■

Proof of Proposition 5.

We first derive the optimal solution under mixed income. Under mixed income, the subgame derivation is the same as in the main model. Thus, we follow the proof of Proposition 1 and write the platform's profit for any given (E,α) as

$$\Pi_P^M=\Pi_P+\rho(1-\theta_{L0})=\frac{\alpha r^*(r^*-1)(8+27r^*-8r^{*2}+48r^{*3})E}{(4r^*-1)^2(8-12r^*+23r^{*2}-4r^{*3})}-\frac{\lambda E^2}{2}+\frac{3r^*\rho}{4r^*-1}, \tag{43}$$

where $r^*(E,\alpha)$ satisfies equation (20). By equations (20) and (43), using the same technique as in the proof of Lemma 2, we obtain the platform's optimal solution under mixed income:

$$\alpha^M=\frac{f(r^*-1)^2(192r^{*4}-80r^{*3}+116r^{*2}+5r^*-8)+\lambda(-4r^{*3}+23r^{*2}-12r^*+8)^2}{C}, \tag{44}$$

$$E^M=\frac{r^*(r^*-1)(4r^*+1)\,(12r^{*2}-5r^*+8)\,(-4r^{*3}+23r^{*2}-12r^*+8)}{(4r^*-1)^2C}, \tag{45}$$

where $C=2f(r^*-1)^2(192r^{*4}-80r^{*3}+116r^{*2}+5r^*-8)+\lambda(-4r^{*3}+23r^{*2}-12r^*+8)^2$ and $r^*(E^M,\alpha^M)$ satisfies equation (20), i.e.,

$$\rho=\frac{r^*(-48r^{*4}+56r^{*3}-35r^{*2}+19r^*+8)g_{15}(r^*)}{3(4r^*-1)^3C^2}. \tag{46}$$

Thus, (E^M, α^M) maximizes Π_P^M with the corresponding $r^M \doteq r^*(E^M, \alpha^M)$. Recall from the proof of Lemma 2, (E^*, α^*) maximizes Π_P with the corresponding $\dot{r}^* = r^*(E^*, \alpha^*)$. We next show that α^M under mixed income ($\rho > 0$) is higher than α^* in the main model ($\rho = 0$). To facilitate the proof, we present the following lemma (proof given later in this appendix).

Lemma A1: *Given the platform's common infrastructure E,*

(i) the optimal commission rate is higher when the platform maximizes its payoff from mixed income ($\rho > 0$) than when it maximizes its payoff from commission income alone ($\rho = 0$).

(ii) its optimal commission rate increases with E under either commission income or mixed income.

There are two cases, which we analyze in turn.

Case 1: $E^M \geq E^*$. We next show this condition implies $\alpha^M \geq \alpha^*$. Lemma A1(ii) shows that for a platform that offers common infrastructure E, the optimal commission rate increases with E under mixed income. This implies that if $E^M \geq E^*$, then $\alpha^M(E^M) \geq \alpha^M(E^*)$. Lemma A1(i) suggests that given E^*, the commission rate under mixed income is higher than that under commission income, i.e., $\alpha^M(E^*) > \alpha^*(E^*)$. Together, we obtain $\alpha^M(E^M) > \alpha^*(E^*)$. Because ($E^M$, α^M) maximizes Π_P^M and (E^*, α^*) maximizes Π_P, we have $\alpha^M = \alpha^M(E^M)$ and $\alpha^* = \alpha^*(E^*)$ so that $\alpha^M > \alpha^*$.

Case 2: $E^M < E^*$. There are two possibilities. a) $\alpha^M \geq \alpha^*$, then no further proof is needed. b) $\alpha^M < \alpha^*$, which never arises as we show below by proof of contradiction.

Suppose $E^M < E^*$ and $\alpha^M < \alpha^*$, by the proof of Lemma 2 we have $r^M = r^*(E^M, \alpha^M) > \dot{r}^* = r^*(E^*, \alpha^*) > 1$ so that $\frac{3\dot{r}^*}{4\dot{r}^*-1} > \frac{3r^M}{4r^M-1} > 0$. Recall that ($E^*$, α^*) maximizes Π_P, we have $\Pi_P(E^*, \alpha^*) \geq \Pi_P(E^M, \alpha^M)$, so that for any $\rho > 0$,

$$\Pi_P(E^*, \alpha^*) + \frac{3\dot{r}^*\rho}{4\dot{r}^* - 1} > \Pi_P(E^M, \alpha^M) + \frac{3r^M\rho}{4r^M - 1} = \Pi_P^M(E^M, \alpha^M),$$

which is a contradiction because (E^M, α^M) maximizes Π_P^M.

Summarizing both cases, we know $\alpha^M \geq \alpha^*$ always holds. Numerical analysis suggests $\frac{\partial E^M}{\partial \rho} > 0$ at $\rho > 0$. We next prove $\frac{\partial E^M}{\partial \rho} > 0$ at $\rho = 0$. By equation (45), we have $\frac{\partial E^M}{\partial \rho} = \frac{\partial E^M}{\partial r^*}\frac{\partial r^*}{\partial \rho}$, where

$$\frac{\partial E^M}{\partial r^*} = \frac{-2fg_{16}(r^*) + \left(-4r^3 + 23r^2 - 12r + 8\right)^2 B\lambda}{(4r^* - 1)^3 C^2}. \tag{47}$$

When $\rho = 0$, equation (46) becomes the same as equation (15). Substituting it into equation (47), we prove for $\underline{r} < r < \bar{r}$, $\frac{\partial E^M}{\partial r^*} < 0$.

Taking derivative with respect to ρ on both sides of equation (46) yields

$$\frac{\partial r^*(E^M, \alpha^M)}{\partial \rho} = \frac{3(4r^* - 1)^4 C^3}{4f^2 g_{17}(r^*) - 2f\lambda g_{18}(r^*) + \lambda^2 g_{19}(r^*)}$$

When $\rho = 0$, substituting equation (15) into it yields

$$\frac{\partial r^*(E^M, \alpha^M)}{\partial \rho}\Big|_{\rho=0} = \frac{-3r^*(4r^* - 1)^4(4r^{*3} - 23r^{*2} + 12r^* - 8)^2 A^2 \lambda}{(r^* - 1)^2(208r^{*3} + 184r^{*2} + 177r^* + 16)g_{20}(r^*)} < 0,$$

because $g_{20}(r^*) > 0$ for $\underline{r} \le r^* \le \bar{r}$ and $\lambda > 0$. Therefore, $\frac{\partial E^M}{\partial \rho} > 0$ at $\rho = 0$. ■

We next establish Lemma A1, which is used in the proof of Proposition 5.

Proof of Lemma A1. (i) The platform's profit under mixed income can be written as $\Pi_P^M = \Pi_P + \rho(D_H + D_L)$. For a platform that offers common infrastructure E, let $\alpha_S^M = \alpha^M(E)$ and $\alpha_S = \alpha^*(E)$ denote its optimal commission rate under mixed income and commission income, respectively. We want to show $\alpha^M(E) \ge \alpha^*(E)$.

Given E, the optimal $\alpha^M(E)$ satisfies the first order condition

$$\frac{\partial \Pi_P^M}{\partial \alpha} = \frac{\partial \Pi_P}{\partial \alpha} - \frac{3\rho}{(4r^* - 1)^2}\frac{\partial r^*}{\partial \alpha} = 0,$$

where $\frac{\partial r^*}{\partial \alpha}$ is given by equation (28). From the proof of Proposition 2, we know that $\frac{\partial r^*}{\partial \alpha} < 0$, so that $\frac{\partial \Pi_P^M}{\partial \alpha} > \frac{\partial \Pi_P}{\partial \alpha}$. From the proof of Lemma 2, we know that $\Pi_P(E, \alpha)$ is concave in α and reaches its maximum at $\alpha^*(E)$. Therefore, $\frac{\partial \Pi_P}{\partial \alpha} \ge 0$ for α in $[0, \alpha^*(E)]$, which implies $\frac{\partial \Pi_P^M}{\partial \alpha} > 0$ always holds in this region so that $\alpha^M(E) > \alpha^*(E)$.

(ii.1) Consider the case of commission income, under which the platform's revenue is $R_P = \alpha(p_H q_H + p_L q_L)$. Given E, Π_P and R_P have the same maximizer $\alpha^*(E)$ because $\Pi_P = R_P - \frac{\lambda E^2}{2}$. We want to show $\frac{\partial \alpha^*(E)}{\partial E} > 0$.

From the proof of Proposition 1, the platform's revenue in stage 1 (based on the vendor's subgame) is

$$R_P = \frac{\alpha r^*(r^* - 1)(1 + 4r^*)(8 - 5r^* + 12r^{*2})E}{(4r^* - 1)^2(8 - 12r^* + 23r^{*2} - 4r^{*3})},$$

where $r^*(E,\alpha)$ satisfies equation (20). Given E, taking derivative of R_P with respect to α and setting it to zero yield

$$\frac{\partial R_P}{\partial \alpha}=\frac{-\alpha r^{*2}B}{f(4r^*-1)^5(-8-40r^*+63r^{*2}-34r^{*3}+64r^{*4})}+\frac{r^*(r^*-1)(1+4r^*)(8-5r^*+12r^{*2})E}{(4r^*-1)^2(8-12r^*+23r^{*2}-4r^{*3})}=0. \tag{48}$$

We prove for $\underline{r}\le r^*\le \bar{r}$, $\frac{\partial^2 R_P}{\partial \alpha^2}=-\frac{2r^{*2}g_{21}(r^*)}{f(4r^*-1)^5\left(-64r^{*4}+4r^{*3}-63r^{*2}+40r^*+8\right)^2B}<0$ because $g_{21}(r^*)>0$ and $B>0$. Therefore, there exists a unique maximizer $\alpha^*(E)$ that satisfies equations (20) and (48), which together yield

$$fE(r^*-1)(4r^*-1)^3A+g_{22}(r^*)=0. \tag{49}$$

Taking derivatives of both sides with respect to E yields

$$\frac{\partial r^*}{\partial E}=\frac{-f(r^*-1)(4r^*-1)^3A}{2(fE(4r^*-1)^2g_{23}(r^*)+g_{24}(r^*))}. \tag{50}$$

From equations (28), (49) and (50), we prove for $\underline{r}\le r^*\le \bar{r}$

$$\frac{\partial \alpha^*}{\partial E}=\frac{\partial \alpha^*}{\partial r^*}\frac{\partial r^*}{\partial E}=\frac{f(r^*-1)(4r^*-1)^3(64r^{*4}-34r^{*3}+63r^{*2}-40r^*-8)AB}{2r^{*2}(-4r^{*3}+23r^{*2}-12r^*+8)g_{25}(r^*)}>0,$$

where $g_{25}(r^*)>0$.

(ii.2) Consider the case of mixed income. Similar as before, we use the platform's revenue under mixed income, $R_P^M=\alpha(p_HD_H+p_LD_L)+\rho(D_H+D_L)$. Given E, Π_P^M and R_P^M have the same maximizer $\alpha^M(E)$.

From the proof of Proposition 1, the platform's revenue (based on the vendors' subgame) is

$$R_P^M=\frac{\alpha r^*(r^*-1)(8+27r^*-8r^{*2}+48r^{*3})E}{(4r^*-1)^2(8-12r^*+23r^{*2}-4r^{*3})}+\frac{3r^*\rho}{4r^*-1}=R_P+\frac{3r^*\rho}{4r^*-1},$$

where $r^*(E,\alpha)$ satisfies equation (20). Taking derivative of R_P^M with respect to α and setting it to zero yield

$$\frac{\partial R_P^M}{\partial \alpha}=\frac{\partial R_P}{\partial \alpha}-\frac{3\rho}{(4r^*-1)^2}\frac{\partial r^*}{\partial \alpha}=0. \tag{51}$$

Substituting $\frac{\partial R_P}{\partial \alpha}$ as in equation (48) and $\frac{\partial r^*}{\partial \alpha}$ as in equation (28) into equation (51) yields

$$\frac{-\alpha E r^{*2} B + 3r^{*2}(4r^*-1)^3(8-12r^*+23r^{*2}-4r^{*3})^2\rho}{fE(4r^*-1)^5(-8-40r^*+63r^{*2}-34r^{*3}+64r^{*4})} + \frac{r^* E(8-5r^*+12r^{*2})(-1-3r^*+4r^{*2})}{(4r^*-1)^2(8-12r^*+23r^{*2}-4r^{*3})} = 0. \quad (52)$$

The optimal commission rate $\alpha^M(E)$ must satisfy both equations (20) and (52). Moreover, from equation (52), we obtain ρ as a function of r^*. Substituting it into R_P^M yields for $\underline{r} \le r^* \le \bar{r}$

$$R_P^M = \frac{E(2fE(4r^*-1)^3(r^*-1)g_{26}(r^*) + r^{*2}(4r^{*3}-23r^{*2}+12r^*-8)g_{27}(r^*))}{(1-4r)^2\left(4r^{*3}-23r^{*2}+12r^*-8\right)^3} \ge 0,$$

where $g_{26}(r^*) > 0$, $g_{27}(r^*) > 0$. Since $R_P^M \ge 0$ holds always, we must have $f \le \hat{f} = -\frac{r^{*2}(4r^{*3}-23r^{*2}+12r^*-8)g_{27}(r^*)}{2E(r^*-1)(4r^*-1)^3 g_{26}(r^*)}$.

Below we prove for $\underline{r} \le r^* \le \bar{r}$,

$$\frac{\partial \alpha^M(E)}{\partial E} = \frac{\partial r^*}{\partial E} \Big/ \frac{\partial r^*}{\partial \alpha} > 0.$$

First, from the proof of Proposition 1, we know $\frac{\partial r^*}{\partial \alpha} < 0$. Next, taking derivatives of both sides of equation (52) with respect to E, we prove that for $0 < f \le \hat{f}$ and $\underline{r} \le r^* \le \bar{r}$

$$\frac{\partial r^*}{\partial E} = \frac{(4r^*-1)\left(4r^{*3}-23r^{*2}+12r^*-8\right)g_{28}(r^*)}{2g_{29}(r^*)E} < 0$$

where $g_{28}(r^*) < 0$ and $g_{29}(r^*) < 0$, and then $\frac{\partial r^*}{\partial E} < 0$ follows. (see supplementary appendix for expressions of g_{28} and g_{29}.) ■